

PART 7

TRADE

17 INTERDEPENDENCE AND THE GAINS FROM TRADE

Consider this typical day. You wake up in the morning and make some coffee from beans grown in Kenya, or tea from leaves grown in Sri Lanka. Over breakfast, you listen to a radio programme on a device made in China. You get dressed in clothes manufactured in Thailand. You drive to the university in a car made of parts manufactured in more than a dozen countries around the world. Then you open up your economics textbook published by a company located in Hampshire in the UK, printed on paper made from trees grown in Finland and written by authors from the United States and England.

Every day you rely on many people from around the world, most of whom you do not know, to provide you with the goods and services that you enjoy. Such interdependence is possible because people trade with one another. One of the early insights from economists like Adam Smith and David Ricardo was that trade can be beneficial, and this insight is something that many economists still hold dear. In this chapter we will look at the benefits to trade and at arguments which cast doubt on the extent to which trade benefits everyone.

THE PRODUCTION POSSIBILITIES FRONTIER

We begin our analysis by looking at a model of the economy where no trade takes place and simplified to the production of goods in two categories, capital goods and consumer goods. The country has a certain amount of resources of land, labour and capital available to allocate between the production of these goods which can be shown using a production possibilities frontier (PPF). (You might also see this referred to as a production possibilities boundary or production possibilities curve – they are the same thing.) The **production possibilities frontier** is a graph that shows the various combinations of output – in this case, capital goods and consumer goods – that the economy can produce given the available factors of production and technology that firms can use to turn these factors into output.

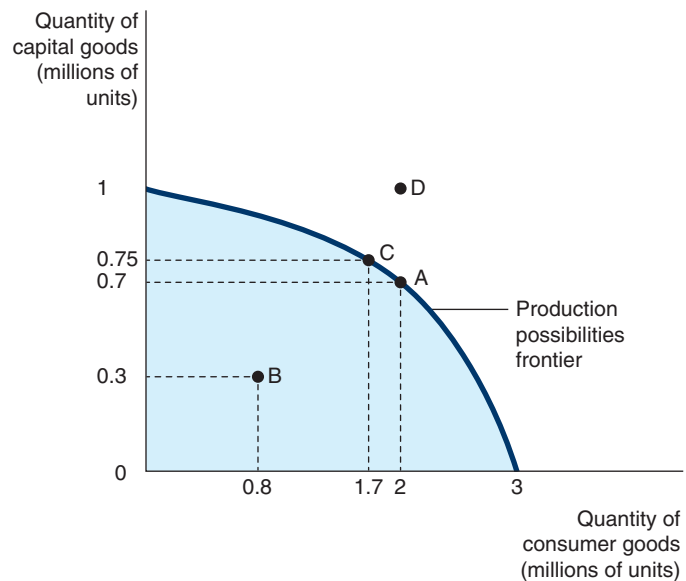
production possibilities frontier a graph that shows the combinations of output that the economy can possibly produce given the available factors of production and technology

Figure 17.1 is an example of a PPF. In this economy, if all resources were devoted to the production of capital goods, the economy would produce 1 million units of capital goods and no consumption goods. If all resources were used to produce consumer goods, the economy would produce 3 million units and no capital goods. The two end points of the PPF represent these extreme possibilities. If the economy were to divide its resources between the two goods, it could produce 700,000 capital goods and 2 million consumer goods, shown in the figure by point A. The economy can produce at any point on or inside the production possibilities frontier, but it cannot produce at points outside the frontier. The outcome at point D is not possible because the economy does not have enough of the factors of production to support that level of output.

FIGURE 17.1

The Production Possibilities Frontier

The production possibilities frontier shows the combinations of output – in this case, capital and consumer goods – that the economy can produce given its factor endowment. The economy can produce any combination on or inside the frontier. Points outside the frontier are not feasible given the economy's existing resources.



An outcome is said to be *efficient* if the economy is getting all it can from the scarce resources it has available. Points on the PPF represent efficient levels of production. When the economy is producing at such a point, say point A, there is no way to produce more of one good without producing less of the other. Point B represents an *inefficient* outcome. For some reason, the economy is producing less than it could from the resources it has available and so some of its resources are lying idle (they are unemployed or underemployed). At point B, the country is producing only 300,000 units of capital goods and 800,000 consumer goods. If the source of the inefficiency were eliminated, the economy could move from point B to point A, increasing production of both capital goods (to 700,000) and consumer goods (to 2 million).

The production possibilities frontier illustrates the idea of a trade-off in that once a society has reached the efficient points on the frontier, the only way of getting more of one good is to get less of the other. When the economy moves from point A to point C, for instance, society produces more capital goods but at the expense of producing fewer consumer goods.

Opportunity Cost The PPF can also be used to show opportunity cost. If some factors of production are reallocated from the consumer goods industry to the capital goods industry, moving the economy from point A to point C, for example, it gives up 300,000 consumer goods to get 50,000 additional capital goods. The opportunity cost of the additional 50,000 capital goods gained is the 300,000 consumer goods sacrificed.

Calculating Opportunity Costs Remember that the opportunity cost is the cost expressed in terms of the next best alternative sacrificed – what must be given up in order to acquire something. In the example the country had to give up 300,000 consumer goods to acquire 50,000 additional capital goods. The opportunity cost can be expressed in terms of either capital goods or consumer goods – they are the reciprocal of each other.

As a general principle we can express the opportunity cost as a ratio expressed as the sacrifice of one good in terms of the gain in the other:

$$\text{Opportunity cost of good } y = \frac{\text{Sacrifice of good } x}{\text{Gain in good } y}$$

Expressing the opportunity cost in terms of good x would give:

$$\text{Opportunity cost of good } x = \frac{\text{Sacrifice of good } Y}{\text{Gain in good } x}$$

The opportunity cost of one additional unit of consumer goods or capital goods can be calculated by firstly writing out the known quantities:

The OC of 300,000 consumer goods is 50,000 capital goods.

Divide both quantities by the number of consumer goods:

$$\text{The OC of } \frac{300,000}{300,000} \text{ consumer goods is } \frac{50,000}{300,000} \text{ capital goods}$$

Now complete the calculation to get the opportunity cost of one additional unit of consumer goods in terms of capital goods sacrificed:

$$\text{The OC of one consumer good is } 0.17 \text{ (2dp) capital goods.}$$

This tells us that for every 1 additional unit of consumer goods acquired, we must give up 0.17 of a capital good.

To find the opportunity cost of capital goods in terms of consumer goods, follow the same process but in reverse.

The OC of 50,000 capital goods is 300,000 consumer goods.

$$\text{The OC of } \frac{50,000}{50,000} \text{ consumer goods is } \frac{300,000}{50,000} \text{ consumer goods}$$

The OC of one capital good is six consumer goods.

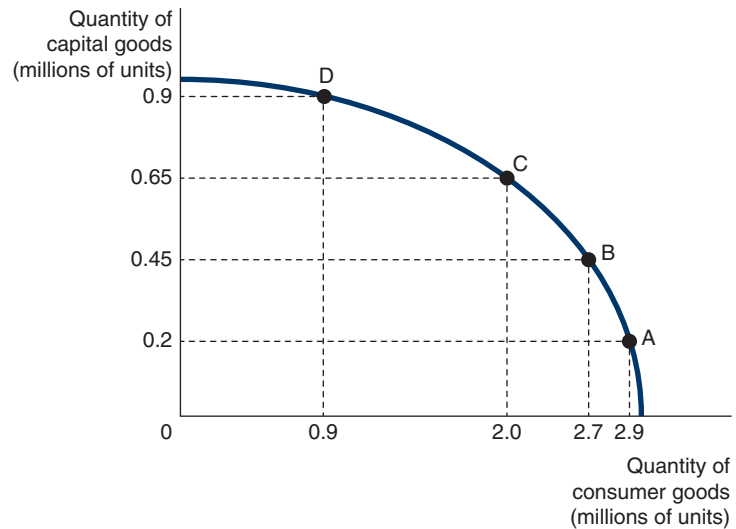
The Shape of the Production Possibilities Frontier

The PPF in Figure 17.2 is bowed outwards (concave to the origin). This is due to the fact that when economies move resources from one use to another, unless they are perfect substitutes (in which case the PPF would be a straight line), as the rate at which the increase in output of one good increases, the opportunity cost in terms of the other changes. For example, if the economy is using most of its resources to make consumer goods at point A in Figure 17.2, land, labour and capital are being used to make consumer goods even if these resources are not best suited to making these goods. If the country moves from point A to point B, the gain in capital goods is 250,000 units but the sacrifice in consumer goods is 200,000. The opportunity cost of one additional unit of capital goods will be 0.8 consumer goods sacrificed. Resources released from producing consumer goods are now able to produce capital goods, a use to which they may be more appropriately suited. If the economy moves from point B to point C, the gain in capital goods is a further 200,000 units but the sacrifice in terms of consumer goods is now 700,000. The opportunity cost of 1 additional unit of capital goods now is 3.5 consumer goods sacrificed.

If the country continues to shift resources to capital goods from consumer goods, the ease of substitutability of factors becomes weaker and the sacrifice in consumer goods becomes greater. Moving from point C to point D yields an additional 250,000 capital goods but at a cost of 1.1 million units of consumer goods. The opportunity cost of 1 additional unit of capital goods is now 4.4 units of consumer goods. The reason that the opportunity cost in terms of consumer goods is rising is that the resources being put to use producing more capital goods are now less suited to the purpose and so the sacrifice in consumer goods increases.

FIGURE 17.2**The Shape of the Production Possibilities Frontier**

The PPF is concave to the origin. The shape of the PPF reflects the opportunity cost of producing different quantities of capital goods and consumer goods. If the country switches resources from consumer goods to capital goods, the opportunity cost in terms of the increase in consumer goods sacrificed for every additional unit of capital goods rises as the output combination moves from point A to point D.



The PPF illustrates two of the key questions any economy must answer: what is to be produced and how will the output be produced? Most economies could use the resources it has at its disposal in a variety of ways. For example, it is possible that the UK could allocate resources to the production of oranges. Large amounts of land could be set aside for the construction of glasshouses in which the climate, water and nutrition needs of orange trees are controlled by computer technology. In Spain the same amount of oranges could be produced using far fewer resources simply because the climate is more conducive to growing oranges. The opportunity cost of using resources in this way in the UK is likely to be high, therefore.

A Shift in the Production Possibilities Frontier

The PPF shows the trade-off between the production of different goods at a given time, but the trade-off can change over time. For example, technological advances can mean that factors of production are considerably more productive in terms of output per unit per time period. Countries might also be in a position to recover or make use of more natural resources, or the effects of education in the country mean the labour force becomes more productive. Over time, therefore, opportunity cost ratios of production change and this can affect the shape and position of the PPF.

Figure 17.3 shows three possible outcomes. In panel (a) the PPF has shifted outwards showing that it is now possible to produce more of both capital goods and consumer goods as indicated by the move to point B. If all resources were devoted to capital goods, the economy could now produce 1.2 million units, and if all resources were devoted to consumer goods, the country could produce 3.6 million units. The relative opportunity cost ratios, however, remain the same because the PPF has shifted outwards, parallel at every point to the original curve. Because of this economic growth, society might move production from point A to point B, enjoying more consumer goods and more capital goods.

Panel (b) shows a shift in the PPF but this time the economic advances in the productivity of the capital goods industry is greater than that of the consumer goods industry. If all resources were now devoted to capital goods production, the country could produce 1.5 million units, and if it devoted all output to consumer goods it could produce 3.2 million units. The opportunity cost ratio at all points on the new PPF will now be different from those on the original PPF.

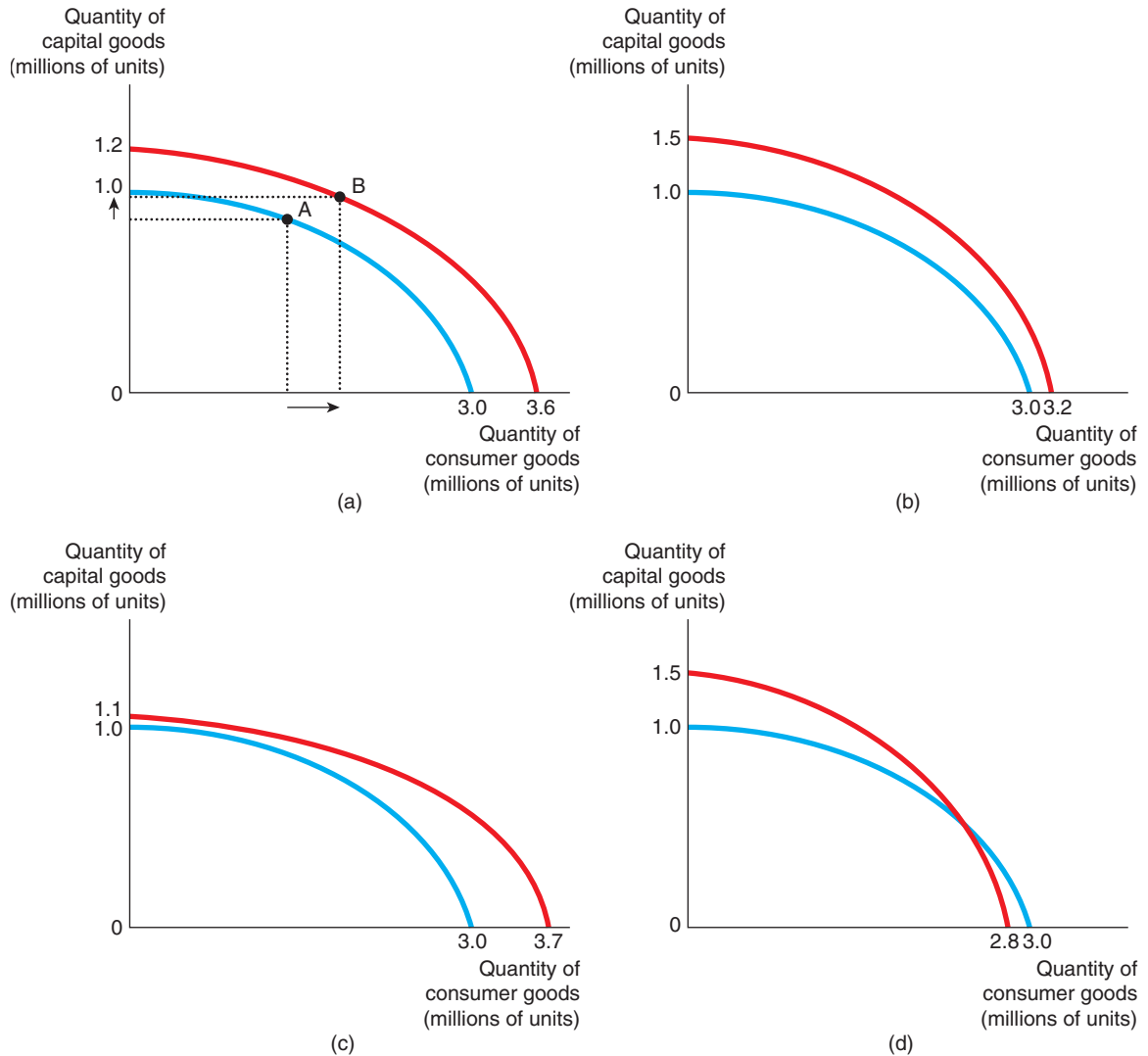
Panel (c) shows the situation where the economic advances in productivity in the consumer goods industry is greater than in the capital goods industry. In this case, if all resources were devoted to producing consumer goods, the country could now produce 3.7 million units, and if all resources were devoted to producing capital goods, 1.1 million units could be produced.

It may also be possible that productivity in one industry might actually reduce while that in the other rises, in which case the PPF might take the look of that in panel (d) of Figure 17.3. In this case there has been an increase in productivity in the capital goods industry but a reduction in the consumer goods industry.

FIGURE 17.3

Shifts in the Production Possibilities Frontier

Panels (a) to (d) show different shifts in the PPF as a result of changes in the productivity of factors used in producing capital and consumer goods.



The PPF simplifies a complex economy to highlight and clarify some basic ideas. We can now extend the analysis to look at how different factor endowments and factor productivity in different countries can lead to countries trading and gaining advantages.

SELF TEST Use the information in Figure 17.3 to calculate the opportunity cost ratios of both capital and consumer goods in each scenario presented in the different panels. Assume, in each case, that the country moves from devoting all its resources to capital goods and then switches to devoting all its resources to consumer goods.

INTERNATIONAL TRADE

Each country has its own PPF and in isolation faces choices about the use of resources to produce goods. If the country wants to increase the amount of goods available to all its citizens, then it can rely on increasing the number of factors it has at its disposal or the efficiency with which its resources are used to shift the PPF outwards. In addition to this, countries might also choose to engage in trade as a means of providing benefits to its citizens and effectively shift the PPF.

A Parable for the Modern Economy

We are going to use a simple example to illustrate how trade can lead to benefits. Imagine that there are two goods in the world, beef and potatoes, and two people in the world (our analogy for two different countries), a cattle farmer named Silvia and a market gardener named Johan, each of whom would like to eat both beef and potatoes.

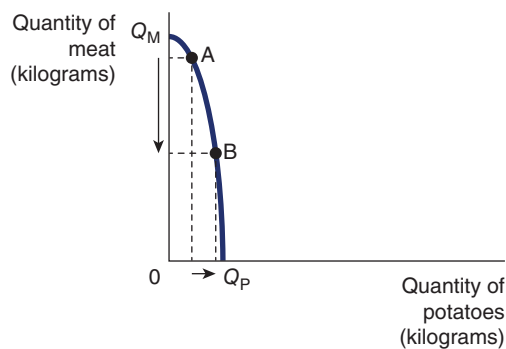
Assume initially that the cattle farmer can produce only meat and the market gardener can produce only potatoes. In one scenario, the farmer and the gardener could choose to have nothing to do with each other. After several months of eating beef roasted, boiled, fried and grilled, the cattle farmer might decide that self-sufficiency is not all she expected. The market gardener, who has been eating potatoes mashed, fried and baked, would most likely agree. It is easy to see that trade would allow them to enjoy greater variety: each could have beef and potatoes.

Now assume that the farmer and the gardener are each capable of producing both goods. In this case each has a PPF analogous to two different countries. Suppose, for example, that the market gardener can rear cattle and produce meat, but that he is not very good at it, and that the cattle farmer is able to grow potatoes, but her land is not very well suited to this crop. The PPF for Johan and Silvia would look like those in Figure 17.4. Panel (a) shows Silvia's PPF and panel (b) shows Johan's. Because Silvia is more efficient in the production of meat than she is producing potatoes, the shape of the PPF reflects the opportunity cost of any decision to divert more of her resources away from producing meat to producing potatoes. If Silvia devoted all her time and resources to producing meat, she could produce Q_M , the vertical intercept. If she makes the decision to divert resources to the production of potatoes, the sacrifice in terms of lost meat output is relatively high compared to the gains in output of potatoes as shown by the move from point A to point B.

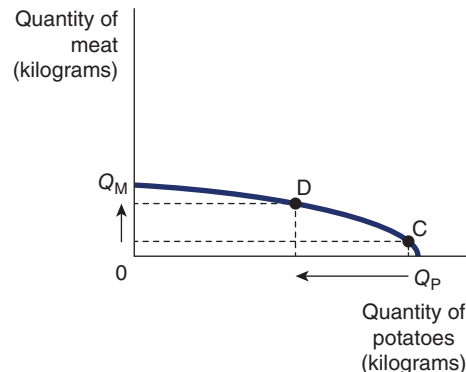
FIGURE 17.4

Differing Opportunity Cost Ratios

Panels (a) and (b) show Silvia and Johan's PPFs respectively. Silvia is more skilled in producing meat while Johan is more skilled in producing potatoes, although both could divert resources to produce the other good. The opportunity cost ratios for each are different – the opportunity cost to Silvia of diverting resources from meat to potatoes is high while for Johan the opportunity cost of diverting resources from potatoes to meat is high.



(a) Silvia's PPF



(b) Johan's PPF

Johan's situation is the reverse of this and is shown in panel (b). If Johan allocates all resources to producing potatoes, he produces Q_p . If he diverts resources away from potato production towards meat production, he sacrifices a relatively large amount of output of potatoes to gain a relatively small amount of meat as shown by the movement from C to D. The opportunity cost of diverting resources to meat for Johan is high.

Economically, it would be more efficient for Johan and Silvia to cooperate with each other, specialize in what they both do best and benefit from trading with each other at some mutually agreeable rate of exchange. For example, they might agree to a rate of exchange of 1 kg of meat for every 5 kg of potatoes.

The gains from trade are less obvious, however, when one person is better at producing *every* good. For example, suppose that Silvia is better at rearing cattle *and* better at growing potatoes than Johan. In this case, should the farmer or gardener choose to remain self-sufficient, or is there still reason for them to trade with each other?

Production Possibilities

Suppose that Johan and Silvia each work 8 hours a day, 6 days a week (a working week of 48 hours) and take Sunday off. They can spend their time growing potatoes, rearing cattle, or a combination of the two. Table 17.1 shows the amount of time each person takes to produce 1 kg of each good. The gardener can produce 1 kg of meat in 6 hours and 1 kg of potatoes in an hour and a half. The farmer, who is more productive in both activities, can produce a kilogram of meat in 2 hours and a kilogram of potatoes in 1 hour. The last columns in Table 17.1 show the amounts of meat or potatoes the gardener and farmer can produce in a 48-hour working week, producing only that good.

TABLE 17.1 The Production Opportunities of Johan the Gardener and Silvia the Farmer

	Time needed to make 1 kg of:		Amount of meat or potatoes produced in 48 hours	
	Meat	Potatoes	Meat	Potatoes
Johan	6 hrs/kg	1.5 hrs/kg	8 kg	32 kg
Silvia	2 hrs/kg	1 hr/kg	24 kg	48 kg

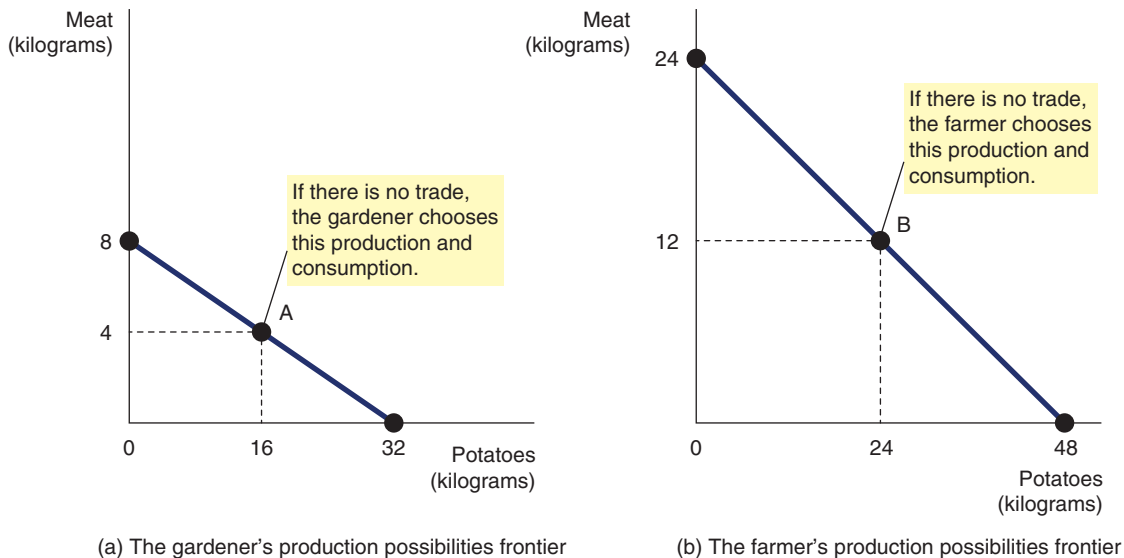
Panel (a) of Figure 17.5 illustrates the amount of meat and potatoes that Johan can produce. If he devotes all 48 hours of his time to potatoes, he produces 32 kg of potatoes (measured on the horizontal axis) and no meat. If he devotes all his time to meat, he produces 8 kg of meat (measured on the vertical axis) and no potatoes. If Johan divides his time equally between the two activities, spending 24 hours a week on each, he produces 16 kg of potatoes and 4 kg of meat. The figure shows these three possible outcomes and all others in between.

This graph is Johan's PPF. Note that in this case, the PPF is a straight line indicating that the slope is constant and thus the opportunity cost to Johan of switching between potatoes and meat is constant. Johan faces a trade-off between producing meat and producing potatoes. If Johan devotes an extra hour to producing meat, he sacrifices potato production. Assume Johan starts at point A producing 4 kg of meat and 16 kg of potatoes. If he then devoted all resources to producing potatoes, he would produce 32 kg of potatoes and sacrifice 4 kg of meat. Using our opportunity cost formula, the opportunity cost of an additional unit of potatoes to Johan is a sacrifice of 4 kg of meat for a gain of 16 kg of potatoes. The opportunity cost of 1 additional kilo of potatoes is, therefore, 0.25 kg of meat. Every additional 1 kg of potatoes produced would involve a trade-off of $\frac{1}{4}$ kg of meat. Conversely, if Johan chose to increase meat production by 1 kg he would have to sacrifice 4 kg of potatoes.

Panel (b) of Figure 17.5 shows the PPF for Silvia. If she devotes all 48 hours of her working week to potatoes, she produces 48 kg of potatoes and no meat. If she devotes all of her time to meat production, she produces 24 kg of meat and no potatoes. If Silvia divides her time equally, spending 24 hours a week on each activity, she produces 24 kg of potatoes and 12 kg of meat. If Silvia moved from devoting half her time producing each to producing all potatoes, the opportunity cost of an additional unit of potatoes is the sacrifice in meat of 12 kg divided by the gain in potatoes of 24 kg. The opportunity cost of an additional unit of potatoes for Silvia is 0.5. Silvia would sacrifice $\frac{1}{2}$ kg of meat for every 1 kg of additional potatoes. The slope of this PPF is, therefore, 0.5. If Silvia shifted production to meat from potatoes, the opportunity cost of an additional 1 kg of meat would be 2 kg of potatoes sacrificed.

FIGURE 17.5**The Production Possibilities Frontier**

Panel (a) shows the combinations of meat and potatoes that Johan can produce. Panel (b) shows the combinations of meat and potatoes that Silvia can produce. Both PPFs are derived from Table 17.1 and the assumption that the gardener and farmer each work 8 hours a day.



If the gardener and farmer choose to be self-sufficient, rather than trade with each other, then each consumes exactly what he or she produces. In this case, the PPF is also the consumption possibilities frontier. That is, without trade, Figure 17.5 shows the possible combinations of meat and potatoes that Johan and Silvia can each consume.

Although these PPFs are useful in showing the trade-offs that the gardener and farmer face, they do not tell us what Johan and Silvia will actually choose to do. To determine their choices, we need to know the tastes of the gardener and the farmer. Assume they choose the combinations identified by points A and B in Figure 17.5: Johan produces and consumes 16 kg of potatoes and 4 kg of meat, while Silvia produces and consumes 24 kg of potatoes and 12 kg of meat.

Specialization and Trade

After several years of feeding her family on combination B, Silvia gets an idea and she goes to talk to Johan:

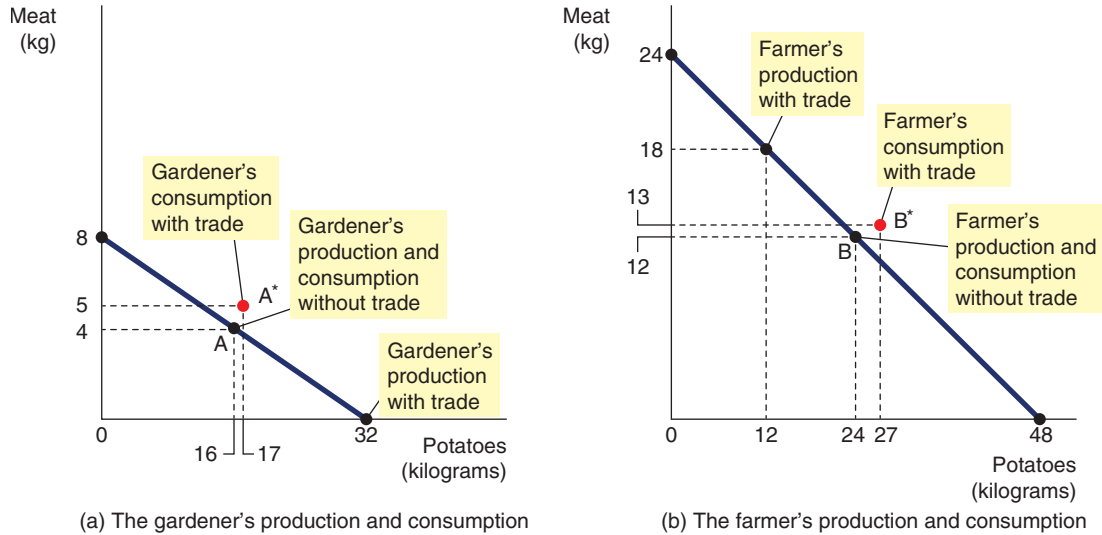
SILVIA: Johan, I have a proposal to put to you. I know how to improve life for both of us. I think you should stop producing meat altogether and devote all your time to growing potatoes. According to my calculations, if you devote all of your working week to growing potatoes, you'll produce 32 kg of potatoes. If you give me 15 of those 32 kg, I'll give you 5 kg of meat in return. You will have 17 kg of potatoes left to enjoy and 5 kg of meat every week, instead of the 16 kg of potatoes and 4 kg of meat you now make do with. If you go along with my plan, you'll have more of *both* foods. (To illustrate her point, Silvia shows Johan panel (a) of Figure 17.6.)

JOHAN: That seems like a good deal for me, Silvia, but how is it that we can both benefit?

SILVIA: Suppose I spend 12 hours a week growing potatoes and 36 hours rearing cattle. Then I can produce 12 kg of potatoes and 18 kg of meat. You will give me 15 kg of your potatoes in exchange for the 5 kg of my meat. This means I end up with 27 kg of potatoes and 13 kg of meat. So I will also be able to consume more of both foods than I do now. (She points out panel (b) of Figure 17.6.)

FIGURE 17.6**How Trade Expands the Set of Consumption Opportunities**

The proposed trade between the gardener and the farmer offers each of them a combination of meat and potatoes that would be impossible in the absence of trade. In panel (a), the gardener consumes at point A* rather than point A. In panel (b), the farmer consumes at point B* rather than point B. Trade allows each to consume more meat and more potatoes.



JOHAN: Hmm, sounds like a good idea, let me think about it.

SILVIA: To help, I've summarized my proposal for you in a simple table. (The farmer hands the gardener a copy of Table 17.2.)

JOHAN: (after pausing to study the table) These calculations seem correct; so we can both be better off?

SILVIA: Yes, because trade allows each of us to specialize in doing what we do best. You will spend more time growing potatoes and less time rearing cattle. I will spend more time rearing cattle and less time growing potatoes. As a result of specialization and trade, each of us can consume more meat and more potatoes without working any more hours.

TABLE 17.2 The Gains from Trade: A Summary

	Johan		Silvia	
	Meat	Potatoes	Meat	Potatoes
Without trade:				
Production and consumption	4 kg	16 kg	12 kg	24 kg
With trade:				
Production	0 kg	32 kg	18 kg	12 kg
With trade	Gets 5 kg	Gives 15 kg	Gives 5 kg	Gets 15 kg
Final consumption	5 kg	17 kg	13 kg	27 kg
Gains from trade:				
Increase in consumption	+1 kg	+1 kg	+1 kg	+3 kg

SELF TEST Draw an example of a PPF for Jan, who is stranded on an island after a shipwreck and spends his time gathering coconuts and catching fish. Does this frontier limit Jan's consumption of coconuts and fish if he lives by himself? Does he face the same limits if he can trade with natives on the island?

THE PRINCIPLE OF COMPARATIVE ADVANTAGE

The principle of *comparative advantage* helps explain why benefits from trade can arise even though Johan, the gardener, is not as efficient in both rearing cattle and growing potatoes as Silvia the farmer.

As a first step in the explanation, consider the following question: in our example, who can produce potatoes at lower cost – the gardener or the farmer? There are two possible answers, which provide the key to understanding the gains from trade. The slope of the production possibilities frontier discussed above will help us to solve the puzzle.

Absolute Advantage

One way to answer the question about the cost of producing potatoes is to compare the inputs required by the two producers. Economists use the term **absolute advantage** when comparing the productivity of one person, firm or nation to that of another. The producer that requires a smaller quantity of inputs to produce a good is said to have an absolute advantage in producing that good.

absolute advantage exists where a producer can produce a good using fewer factor inputs than another

In our example, the farmer has an absolute advantage both in producing meat and in producing potatoes, because she requires less time than the gardener to produce a unit of either good. The farmer needs to input only 2 hours to produce a kilogram of meat, whereas the gardener needs 6 hours. Similarly, Silvia needs only 1 hour to produce a kilogram of potatoes, whereas Johan needs 1.5 hours. Based on this information, we can conclude that the farmer has the lower cost of producing potatoes, if we measure cost in terms of the quantity of inputs.

Opportunity Cost and Comparative Advantage

There is another way to look at the cost of producing potatoes. Rather than comparing inputs required, we can compare the opportunity costs. Let’s first consider Silvia’s opportunity cost in relation to the amount of hours she needs to work. According to Table 17.1, producing 1 kg of potatoes takes her 1 hour of work. When Silvia spends that 1 hour producing potatoes, she spends 1 hour less producing meat. Because Silvia needs 2 hours to produce 1 kg of meat, 1 hour of work would yield $\frac{1}{2}$ kg of meat. Hence the farmer’s opportunity cost of producing 1 kg of potatoes is $\frac{1}{2}$ kg of meat.

Now consider Johan’s opportunity cost. Producing 1 kg of potatoes takes him $1\frac{1}{2}$ hours. Because he needs 6 hours to produce 1 kg of meat, $1\frac{1}{2}$ hours of work would yield $\frac{1}{4}$ kg of meat. Hence the gardener’s opportunity cost of 1 kg of potatoes is $\frac{1}{4}$ kg of meat. These are the opportunity costs we worked out using our formula earlier.

Table 17.3 shows the opportunity costs of meat and potatoes for the two producers. Remember that the opportunity cost of meat is the inverse of the opportunity cost of potatoes. Because 1 kg of potatoes costs the farmer $\frac{1}{2}$ kg of meat, 1 kg of meat costs the farmer 2 kg of potatoes. Similarly, because 1 kg of potatoes costs the gardener $\frac{1}{4}$ kg of meat, 1 kg of meat costs the gardener 4 kg of potatoes.

TABLE 17.3 The Opportunity Cost of Meat and Potatoes

	Opportunity cost of:	
	1 kg of meat	1 kg of potatoes
Gardener	4 kg potatoes	0.25 kg meat
Farmer	2 kg potatoes	0.5 kg meat

Comparative advantage describes the opportunity cost of two producers; the producer who gives up less of other goods to produce good X has the smaller opportunity cost of producing good X and is said to have a comparative advantage in producing it. In our example, Johan the gardener has a lower opportunity cost of producing potatoes than does Silvia, the farmer: a kilogram of potatoes costs Johan only $\frac{1}{4}$ kg of meat, while it costs Silvia $\frac{1}{2}$ kg of meat. Conversely, Silvia has a lower opportunity cost of producing meat than does Johan: a kilogram of meat costs Silvia 2 kg of potatoes, while it costs Johan 4 kg of potatoes. Thus, Johan the gardener has a comparative advantage in growing potatoes, and Silvia the farmer has a comparative advantage in producing meat.

comparative advantage the comparison among producers of a good according to their opportunity cost. A producer is said to have a comparative advantage in the production of a good if the opportunity cost is lower than that of another producer

Although it is possible for one person to have an absolute advantage in both goods (as Silvia does in our example), it is impossible for one person to have a comparative advantage in both goods. Because the opportunity cost of one good is the inverse of the opportunity cost of the other, if a person's opportunity cost of one good is relatively high, their opportunity cost of the other good must be relatively low. Comparative advantage reflects the relative opportunity cost. Unless two people have exactly the same opportunity cost, one person will have a comparative advantage in one good, and the other will have a comparative advantage in the other good.

Comparative Advantage and Trade

In theory, differences in opportunity cost and comparative advantage create the gains from trade. The theory predicts that when each person specializes in producing the good for which they have a comparative advantage, total production in the economy rises, and this increase in the size of the economic cake can be used to make everyone better off.

Consider the proposed deal from the viewpoint of Johan. He gets 5 kg of meat in exchange for 15 kg of potatoes. In other words, Johan buys each kilogram of meat for a price of 3 kg of potatoes. This price of meat is lower than his opportunity cost for 1 kg of meat, which is 4 kg of potatoes. Thus, the gardener benefits from the deal because he gets to buy meat at a good price.

Now consider the deal from Silvia's viewpoint. The farmer buys 15 kg of potatoes for a price of 5 kg of meat. That is, the price of potatoes is $\frac{1}{3}$ kg of meat. This price of potatoes is lower than her opportunity cost of 1 kg of potatoes, which is $\frac{1}{2}$ kg of meat. The farmer benefits because she can buy potatoes at a good price.

These benefits arise because each person concentrates on the activity for which they have the lower opportunity cost: the gardener spends more time growing potatoes, and the farmer spends more time producing meat. As a result, the total production of potatoes and the total production of meat both rise. In our example, potato production rises from 40 to 44 kg, and meat production rises from 16 to 18 kg. The gardener and farmer share the benefits of this increased production.

FYI



The Legacy of Adam Smith and David Ricardo

Economists have long understood the principle of comparative advantage. Here is how Adam Smith put the argument:

It is the maxim of every prudent master of a family, never to attempt to make at home what it will cost him [sic] more to make than to buy. The tailor does not attempt to make his own shoes, but buys them off the shoemaker. The

(Continued)

shoemaker does not attempt to make his own clothes but employs a tailor. The gardener attempts to make neither the one nor the other, but employs those different artificers. All of them find it for their interest to employ their whole industry in a way in which they have some advantage over their neighbours, and to purchase with a part of its produce, or what is the same thing, with the price of part of it, whatever else they have occasion for.

This quotation is from Smith's 1776 book, *An Inquiry into the Nature and Causes of the Wealth of Nations*, which was a landmark in the analysis of trade and economic interdependence.

Smith's book inspired David Ricardo to become an economist, having already made his fortune as a stockbroker in the City of London. In his 1817 book, *Principles of Political Economy and Taxation*, Ricardo developed the principle of comparative advantage as we know it today. The principle was originally put forward by Robert Torrens, a British Army officer and owner of the *Globe* newspaper, in 1815. Ricardo's defence of free trade was not a mere academic exercise. Ricardo put his economic beliefs to work as a member of the British parliament, where he opposed the Corn Laws, which restricted the import of grain.

The conclusions of Adam Smith and David Ricardo form the basis of arguments in favour of free trade and against the imposition of tariffs or other restraints on trade. These arguments have had their critics, including the late Joan Robinson, a widely respected Cambridge economist, who noted that the theory Ricardo developed was largely based on an historical context of early nineteenth-century Britain and its trade with Portugal, and that economies had evolved over time. Some of the assumptions that Ricardo had made do not hold in modern economies in the same way (that all resources are employed and that prices are stable), and this means that the theory must be reassessed in the light of the way economies have changed.

SELF TEST Jan can gather 10 coconuts or catch one fish per hour. His friend, Marie, can gather 30 coconuts or catch two fish per hour. What is Jan's opportunity cost of catching one fish? What is Marie's? Who has an absolute advantage in catching fish? Who has a comparative advantage in catching fish?

Should Countries in Europe Trade with Other Countries?

Our model of Johan and Silvia can be extended to represent whole countries. Many of the goods that Europeans enjoy are produced abroad, and many of the goods produced across Europe are sold abroad. Goods produced abroad and purchased for use in the domestic economy are called **imports**. An import leads to a flow of money out of the country in payment. Goods produced domestically and sold abroad are called **exports** and lead to a flow of money into the country.

imports goods produced abroad and purchased for use in the domestic economy leading to an outflow of funds from a country

exports goods produced domestically and sold abroad leading to an inflow of funds into a country

To see how countries can benefit from trade, suppose there are two countries, Germany and the Netherlands, and two goods, machine tools and cut flowers. Imagine that the two countries produce cut flowers equally well: a German worker and a Dutch worker can each produce 1 tonne per month. By contrast, because Germany has more land suitable for manufacturing, it is better at producing machine tools: a German worker can produce 2 tonnes of machine tools per month, whereas a Dutch worker can produce only 1 tonne of machine tools per month.

The principle of comparative advantage states that each good should be produced by the country that has the smaller opportunity cost of producing that good. Because the opportunity cost of an additional 1 tonne of cut flowers is 2 tonnes of machine tools in Germany but only 1 tonne of machine tools in the Netherlands, the Dutch have a comparative advantage in producing cut flowers. The Netherlands should produce more cut flowers than it wants for its own use and export some of them to Germany. Similarly,

because the opportunity cost of a tonne of cut flowers is 1 tonne of machine tools in the Netherlands but only $\frac{1}{2}$ a tonne of machine tools in Germany, the Germans have a comparative advantage in producing machine tools. Germany should produce more machine tools than it wants to consume and export some of it to the Netherlands. Through specialization and trade, both countries can have more machine tools and more cut flowers.

In reality, of course, the issues involved in trade among nations are more complex than this simple example suggests. Most important among these issues is that each country has many citizens with different interests. International trade can make some individuals worse off, even as it makes the country as a whole better off. When Germany exports machine tools and imports cut flowers, the impact on a German machine tools worker is not the same as the impact on a German cut flower worker.

SELF TEST Suppose that the world's fastest typist happens to be trained in brain surgery. Should they do their own typing or hire a personal assistant? Explain.

THE DETERMINANTS OF TRADE

Having seen that there are benefits to countries of trading, in this next section we look at the gains and losses of international trade. Consider the market for olive oil. The olive oil market is well suited to examining the gains and losses from international trade: olive oil is made in many countries around the world, and there is much world trade in it. Moreover, the olive oil market is one in which policymakers often consider (and sometimes implement) trade restrictions to protect domestic olive oil producers from foreign competitors. We examine here the olive oil market in the imaginary country of Isoland.

The Equilibrium without Trade

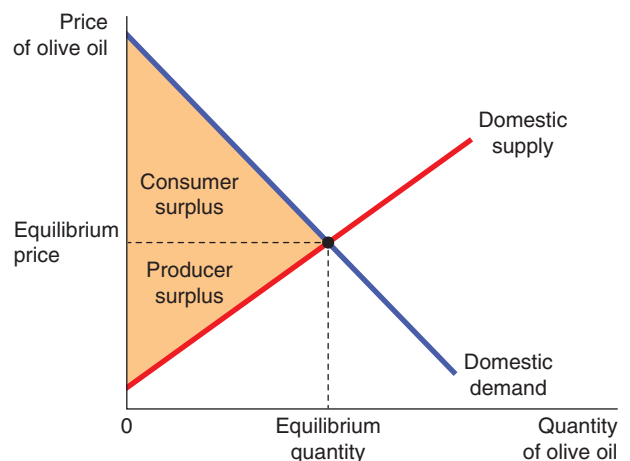
Assume that the Isolandian olive oil market is isolated from the rest of the world. By government decree, no one in Isoland is allowed to import or export olive oil, and the penalty for violating the decree is so large that no one dares try.

Because there is no international trade, the market for olive oil in Isoland consists solely of Isolandian buyers and sellers. As Figure 17.7 shows, the domestic price adjusts to balance the quantity supplied by domestic sellers and the quantity demanded by domestic buyers. The figure shows the consumer and producer surplus in equilibrium without trade. The sum of consumer and producer surplus measures the total benefits that buyers and sellers receive from the olive oil market.

FIGURE 17.7

The Equilibrium without International Trade

When an economy cannot trade in world markets, the price adjusts to balance domestic supply and demand. This figure shows consumer and producer surplus in an equilibrium without international trade for the olive oil market in the imaginary country of Isoland.



Now suppose that Isoland elects a new president. The president campaigned on a platform of 'change' and promised the voters bold new ideas. Their first act is to assemble a team of economists to evaluate Isolandian trade policy. The president asks them to report back on three questions:

- If the government allowed Isolandians to import and export olive oil, what would happen to the price of olive oil and the quantity of olive oil sold in the domestic olive oil market?
- Who would gain from free trade in olive oil and who would lose, and would the gains exceed the losses?
- Should a tariff (a tax on olive oil imports) or an import quota (a limit on olive oil imports) be part of the new trade policy?

The World Price and Comparative Advantage

The first issue our economists take up is whether Isoland is likely to become an olive oil importer or an olive oil exporter. In other words, if free trade were allowed, would Isolandians end up buying or selling olive oil in world markets? To answer this question, the economists compare the current Isolandian price of olive oil with the price of olive oil in other countries. We call the price prevailing in world markets the **world price**. If the world price of olive oil is higher than the domestic price, then Isoland would become an exporter of olive oil once trade is permitted. Isolandian olive oil producers would be eager to receive the higher prices available abroad and would start selling their olive oil to buyers in other countries. Conversely, if the world price of olive oil is lower than the domestic price, then Isoland would become an importer of olive oil. Because foreign sellers offer a better price, Isolandian olive oil consumers would buy olive oil from other countries.

world price the price of a good that prevails in the world market for that good

In essence, comparing the world price and the domestic price before trade indicates whether Isoland has a comparative advantage in producing olive oil. The domestic price reflects the opportunity cost of olive oil: it tells us how much an Isolandian must give up to get one unit of olive oil. If the domestic price is low, the cost of producing olive oil in Isoland is low, suggesting that Isoland has a comparative advantage in producing olive oil relative to the rest of the world. If the domestic price is high, then the cost of producing olive oil in Isoland is high, suggesting that foreign countries have a comparative advantage in producing olive oil. By comparing the world price and the domestic price before trade, we can determine whether Isoland is better or worse at producing olive oil than the rest of the world.

THE WINNERS AND LOSERS FROM TRADE

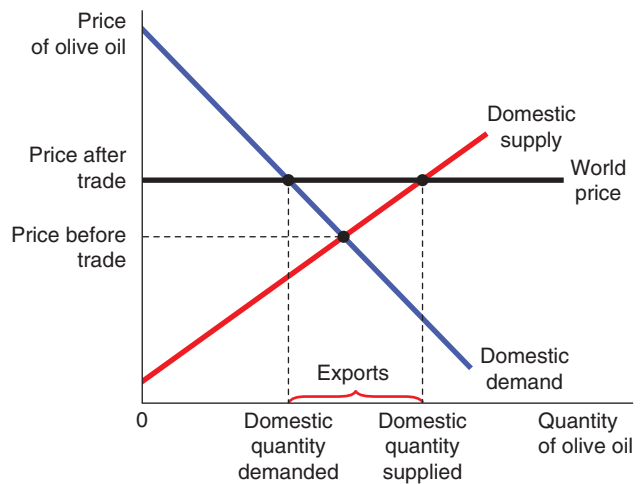
To analyze the welfare effects of free trade, the Isolandian economists begin with the assumption that Isoland is a small economy compared with the rest of the world, so that its actions have a negligible effect on world markets. The small economy assumption has a specific implication for analyzing the olive oil market: if Isoland is a small economy, then the change in Isoland's trade policy will not affect the world price of olive oil. The Isolandians are said to be *price-takers* in the world economy. They can sell olive oil at this price and be exporters or buy olive oil at this price and be importers.

The Gains and Losses of an Exporting Country

Figure 17.8 shows the Isolandian olive oil market when the domestic equilibrium price before trade is below the world price. Once free trade is allowed, the domestic price rises to equal the world price. No seller of olive oil would accept less than the world price, and no buyer would pay more than the world price.

FIGURE 17.8**International Trade in an Exporting Country**

Once trade is allowed, the domestic price rises to equal the world price. The supply curve shows the quantity of olive oil produced domestically, and the demand curve shows the quantity consumed domestically. Exports from Isoland equal the difference between the domestic quantity supplied and the domestic quantity demanded at the world price.



With the domestic price now equal to the world price, the domestic quantity supplied is greater than the domestic quantity demanded, Isoland sells olive oil to other countries. Thus, Isoland becomes an olive oil exporter. Although domestic quantity supplied and domestic quantity demanded differ, the olive oil market is still in equilibrium because there is now another participant in the market: the rest of the world. One can view the horizontal line at the world price as representing the demand for olive oil from the rest of the world. This demand curve is perfectly price elastic because Isoland, as a small economy, can sell as much olive oil as it wants at the world price.

Now consider the gains and losses from opening up trade. Clearly, not everyone benefits. Trade forces the domestic price to rise to the world price. Domestic producers of olive oil are better off because they can now sell olive oil at a higher price, but domestic consumers of olive oil are worse off because they must buy olive oil at a higher price.

To measure these gains and losses, we look at the changes in consumer and producer surplus, which are shown in the graph and table in Figure 17.9. Before trade is allowed, the price of olive oil adjusts to balance domestic supply and domestic demand. Consumer surplus, the area between the demand curve and the before trade price, is area $A + B$. Producer surplus, the area between the supply curve and the before trade price, is area C . Total surplus before trade, the sum of consumer and producer surplus, is area $A + B + C$.

After trade is allowed, the domestic price rises to the world price. Consumer surplus is area A (the area between the demand curve and the world price). Producer surplus is area $B + C + D$ (the area between the supply curve and the world price). Thus, total surplus with trade is area $A + B + C + D$.

These welfare calculations show who wins and who loses from trade in an exporting country, given the assumptions of the model. Sellers benefit because producer surplus increases by the area $B + D$. Buyers are worse off because consumer surplus decreases by the area B . Because the gains of sellers exceed the losses of buyers by the area D , total surplus in Isoland increases.

This analysis of an exporting country yields two conclusions:

- When a country allows trade and becomes an exporter of a good, domestic producers of the good are better off and domestic consumers of the good are worse off.
- Trade raises the economic well-being of a nation in the sense that the gains of the winners exceed the losses of the losers.

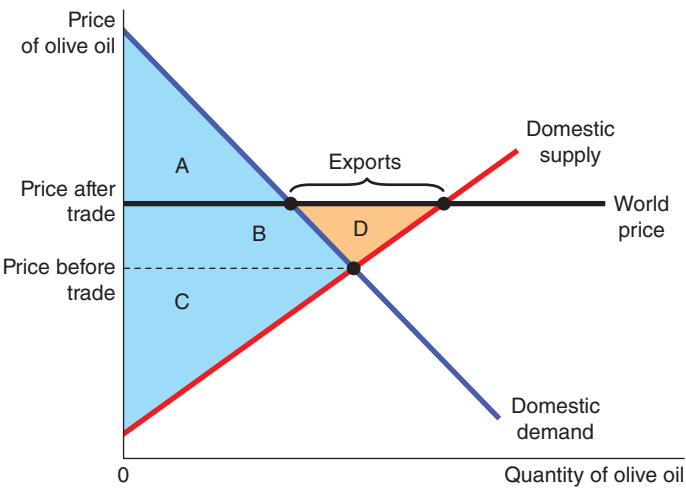
FIGURE 17.9

How Free Trade Affects Welfare in an Exporting Country

When the domestic price rises to equal the world price, sellers are better off (producer surplus rises from C to $B + C + D$), and buyers are worse off (consumer surplus falls from $A + B$ to A). Total surplus rises by an amount equal to area D , indicating that trade raises the economic well-being of the country as a whole.

	Before trade	After trade	Change
Consumer surplus	$A + B$	A	$-B$
Producer surplus	C	$B + C + D$	$+(B + D)$
Total surplus	$A + B + C$	$A + B + C + D$	$+D$

The area D shows the increase in total surplus and represents the gains from trade.



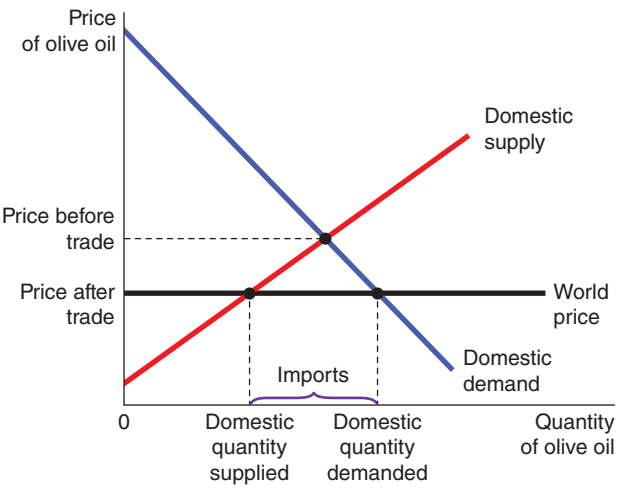
The Gains and Losses of an Importing Country

Now suppose that the domestic price before trade is above the world price. Once again, after free trade is allowed, the domestic price must equal the world price. As Figure 17.10 shows, the domestic quantity supplied is less than the domestic quantity demanded. The difference between the domestic quantity demanded and the domestic quantity supplied is bought from other countries, and Isoland becomes an olive oil importer.

FIGURE 17.10

International Trade in an Importing Country

Once trade is allowed, the domestic price falls to equal the world price. The supply curve shows the amount produced domestically, and the demand curve shows the amount consumed domestically. Imports equal the difference between the domestic quantity demanded and the domestic quantity supplied at the world price.



In this case, the horizontal line at the world price represents the supply of the rest of the world. This supply curve is perfectly elastic because of the assumptions made that Isoland is a small economy and, therefore, can buy as much olive oil as it wants at the world price.

Now consider the gains and losses from trade. Once again, not everyone benefits. When trade forces the domestic price to fall, domestic consumers are better off (they can now buy olive oil at a lower price), and domestic producers are worse off (they now must sell olive oil at a lower price). Changes in consumer and producer surplus measure the size of the gains and losses, as shown in the graph and table in Figure 17.11. Before trade, consumer surplus is area A, producer surplus is area B + C, and total surplus is area A + B + C. After trade is allowed, consumer surplus is area A + B + D, producer surplus is area C, and total surplus is area A + B + C + D.

These welfare calculations show who wins and who loses from trade in an importing country, given the assumptions of the model. Buyers benefit because consumer surplus increases by the area B + D. Sellers are worse off because producer surplus falls by the area B. The gains of buyers exceed the losses of sellers, and total surplus increases by the area D.

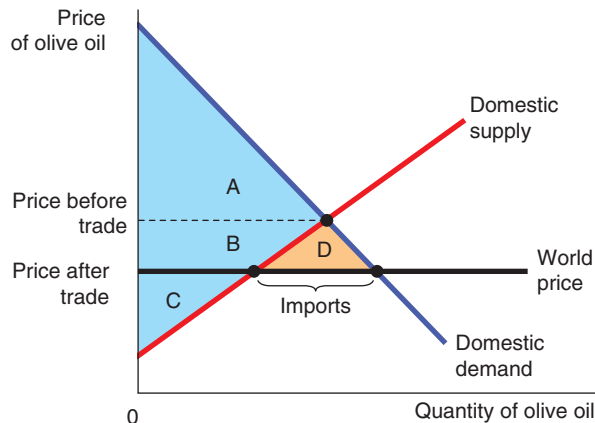
FIGURE 17.11

How Free Trade Affects Welfare in an Importing Country

When the domestic price falls to equal the world price, buyers are better off (consumer surplus rises from A to A + B + D), and sellers are worse off (producer surplus falls from B + C to C). Total surplus rises by an amount equal to area D, indicating that trade raises the economic well-being of the country as a whole.

	Before trade	After trade	Change
Consumer surplus	A	A + B + D	+(B + D)
Producer surplus	B + C	C	-B
Total surplus	A + B + C	A + B + C + D	+D

The area D shows the increase in total surplus and represents the gains from trade.



This analysis of an importing country yields two conclusions parallel to those for an exporting country:

- When a country allows trade and becomes an importer of goods, domestic consumers of the good are better off and domestic producers of the good are worse off.
- Trade raises the economic well-being of a nation in the sense that the gains of the winners exceed the losses of the losers.

Having completed our analysis of trade, we can draw a conclusion that trade can make everyone better off. If Isoland opens up its olive oil market to international trade, the change will create winners and losers, regardless of whether Isoland ends up exporting or importing olive oil. Notice that in our analysis, we have not made a judgement about the winners and losers – whether the gain to the producers is more valuable than the loss to the consumers. In this analysis the key is the effect on total welfare, which in this case has risen for Isoland. In the real world, policymakers may have to take into consideration the power which resides with different groups. If domestic consumers of olive oil in Isoland had considerable lobbying power compared with olive oil producers, then policy decisions may be affected which distort outcomes and reduce total welfare. The effect on consumers, for example, might be limited in comparison with the gains to producers, but presenting arguments in this way does not always win political points! This is something that must always be considered, because while economic analysis may point to a clear policy decision and outcome, there are many other factors that decision-makers must take into account, as exemplified when we look at the arguments for restricting trade.

In our example, the gains of the winners exceed the losses of the losers, so the winners could compensate the losers and still be better off. In this sense, trade *can* make everyone better off. But *will* trade make everyone better off? Probably not.

In practice, compensation for the losers from international trade is rare. Without such compensation, opening up to international trade is a policy that expands the size of the economic cake, while perhaps leaving some participants in the economy with a smaller slice.

We can now see why the debate over trade policy is so often contentious. Whenever a policy creates winners and losers, the stage is set for a political battle. Nations sometimes fail to enjoy the gains from trade simply because the losers from free trade have more political influence than the winners. The losers lobby for trade restrictions, such as tariffs and import quotas.

FYI



Other Benefits of International Trade

Our conclusions so far have been based on the standard analysis of international trade. There are several other economic benefits of trade beyond those emphasized in the standard analysis which can be taken into account. Here, in a nutshell, are some of these other benefits:

- **Increased variety of goods.** Goods produced in different countries are not exactly the same. German beer, for instance, is not the same as US beer. Free trade gives consumers in all countries greater variety from which to choose.
- **Lower unit costs through economies of scale.** Some goods can be produced at low unit or average cost only if they are produced in large quantities. A firm in a small country cannot take full advantage of economies of scale if it can sell only in a small domestic market. Free trade gives firms access to larger world markets and allows them to realize economies of scale more fully.
- **Increased competition.** A company shielded from foreign competitors is more likely to have market power, which in turn gives it the ability to raise prices above competitive levels. This is a type of market failure. Opening up trade fosters competition with the benefits that arise from more competitive markets.
- **Enhanced flow of ideas.** The transfer of technological advances around the world is often thought to be linked to international trade in the goods that embody those advances. The best way for a poor, agricultural nation to learn about the computer revolution, for instance, is to buy some computers from abroad, rather than trying to make them domestically.
- **Generates economic growth.** For poor countries, the increase in output can be a trigger to generating economic growth, which may also bring an improvement in the standard of living for its citizens.

RESTRICTIONS ON TRADE

Despite the benefits that can arise from trade, the fact that there will always be winners and losers as we have seen means that arguments for restricting trade in some way are regularly promoted. We will look at three main methods of restricting trade – tariffs, quotas and non-tariff barriers – and then some of the arguments for trade barriers.

The Effects of a Tariff

The Isolandian economists turn their attention to considering the effects of a **tariff** – a tax on imported goods. The tariff matters only if Isoland becomes an olive oil importer. Concentrating their attention on this case, the economists compare welfare with and without the tariff.

tariff a tax on goods produced abroad and sold domestically

The graph in Figure 17.12 shows the Isolandian market for olive oil. Under free trade, the domestic price equals the world price. A tariff raises the price of imported olive oil above the world price by the amount of the tariff. Domestic suppliers of olive oil, who compete with suppliers of imported olive oil, can now sell their olive oil for the world price plus the amount of the tariff. Thus, the price of olive oil – both imported and domestic – rises by the amount of the tariff and is, therefore, closer to the price that would prevail without trade.

The change in price affects the behaviour of domestic buyers and sellers. Because the tariff raises the price of olive oil, it reduces the domestic quantity demanded from Q_1^D to Q_2^D and raises the domestic quantity supplied from Q_1^S to Q_2^S . Thus, the tariff reduces the quantity of imports and moves the domestic market closer to its equilibrium without trade.

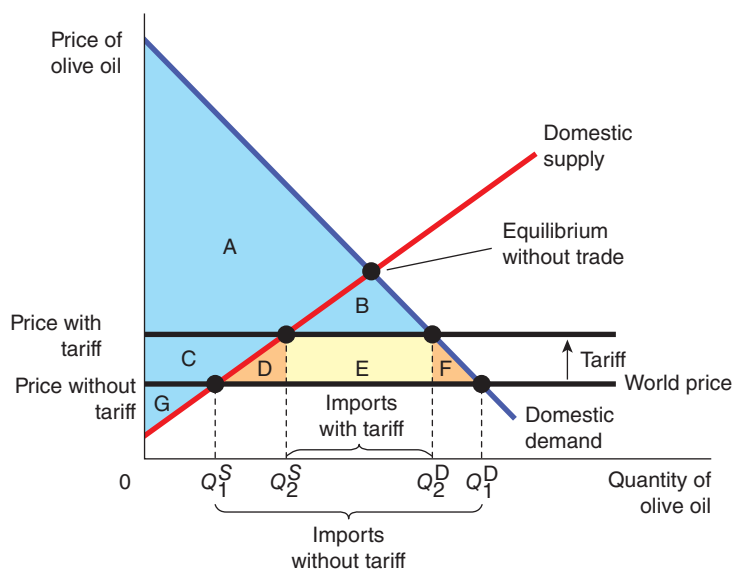
Now consider the gains and losses from the tariff. Because the tariff raises the domestic price, domestic sellers are better off and domestic buyers are worse off. In addition, the government raises revenue. To measure these gains and losses, we look at the changes in consumer surplus, producer surplus and government revenue. These changes are summarized in the table in Figure 17.12.

FIGURE 17.12**The Effects of a Tariff**

A tariff reduces the quantity of imports and moves a market closer to the equilibrium that would exist without trade. Total surplus falls by an amount equal to area $D + F$. These two triangles represent the deadweight loss from the tariff.

	Before tariff	After tariff	Change
Consumer surplus	$A + B + C + D + E + F$	$A + B$	$-(C + D + E + F)$
Producer surplus	G	$C + G$	$+C$
Government revenue	None	E	$+E$
Total surplus	$A + B + C + D + E + F + G$	$A + B + C + E + G$	$-(D + F)$

The area $D + F$ shows the fall in total surplus and represents the deadweight loss of the tariff.



Before the tariff, the domestic price equals the world price. Consumer surplus, the area between the demand curve and the world price, is area $A + B + C + D + E + F$. Producer surplus, the area between the supply curve and the world price, is area G . Government revenue equals zero. Total surplus – the sum of consumer surplus, producer surplus and government revenue – is area $A + B + C + D + E + F + G$.

Once the government imposes a tariff, the domestic price exceeds the world price by the amount of the tariff. Consumer surplus is now area $A + B$. Producer surplus is area $C + G$. Government revenue, which is the quantity of after tariff imports times the size of the tariff, is area E . Thus, total surplus with the tariff is area $A + B + C + E + G$.

To determine the total welfare effects of the tariff, we add the change in consumer surplus (which is negative), the change in producer surplus (positive) and the change in government revenue (positive). We find that total surplus in the market decreases by the area $D + F$. This fall in total surplus is the *deadweight loss* of the tariff.

A tariff causes a deadweight loss simply because a tariff is a type of tax and can distort incentives and change the allocation of resources. In this case, we can identify two effects. First, the tariff on olive oil raises the price of olive oil that domestic producers can charge above the world price and, as a result, encourages them to increase production of olive oil (from Q_1^S to Q_2^S). Second, the tariff raises the price that domestic olive oil buyers must pay and, therefore, encourages them to reduce consumption of olive oil (from Q_1^D to Q_2^D). Area D represents the deadweight loss from the over-production of olive oil, and area F represents the deadweight loss from the under-consumption. The total deadweight loss of the tariff is the sum of these two triangles.

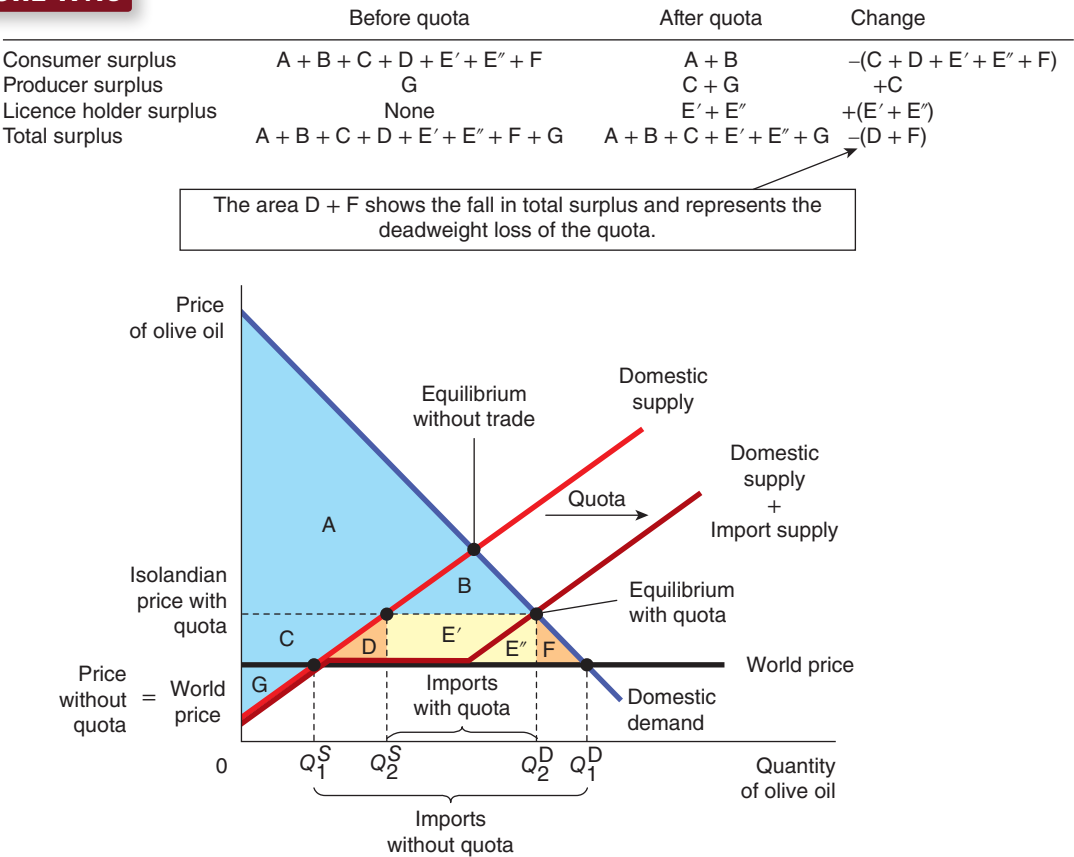
The Effects of an Import Quota

The Isolandian economists next consider the effects of an **import quota** – a limit on the quantity of imports. Imagine that the Isolandian government distributes a limited number of import licences. Each licence gives the licence holder the right to import 1 tonne of olive oil into Isoland from abroad. The Isolandian economists want to compare welfare under a policy of free trade and welfare with the addition of this import quota.

import quota a limit on the quantity of a good that can be produced abroad and sold domestically

The graph and table in Figure 17.13 show how an import quota affects the Isolandian market for olive oil. Because the import quota prevents Isolandians from buying as much olive oil as they want from abroad, the supply of olive oil is no longer perfectly elastic at the world price. Instead, as long as the price of olive oil

FIGURE 17.13



The Effects of an Import Quota

An import quota, like a tariff, reduces the quantity of imports and moves a market closer to the equilibrium that would exist without trade. Total surplus falls by an amount equal to area $D + F$. These two triangles represent the deadweight loss from the quota. In addition, the import quota transfers $E' + E''$ to whoever holds the import licences.

in Isoland is above the world price, the licence holders import as much as they are permitted, and the total supply of olive oil in Isoland equals the domestic supply plus the quota amount. That is, the supply curve above the world price is shifted to the right by exactly the amount of the quota. The supply curve below the world price does not shift because, in this case, importing is not profitable for the licence holders.

The price of olive oil in Isoland adjusts to balance supply (domestic plus imported) and demand. As the figure shows, the quota causes the price of olive oil to rise above the world price. The domestic quantity demanded falls from Q_1^D to Q_2^D and the domestic quantity supplied rises from Q_1^S to Q_2^S . Not surprisingly, the import quota reduces olive oil imports.

Because the quota raises the domestic price above the world price, domestic sellers are better off and domestic buyers are worse off. In addition, the licence holders are better off, because they make a profit from buying at the world price and selling at the higher domestic price. To measure these gains and losses, we look at the changes in consumer surplus, producer surplus and licence holder surplus.

Before the government imposes the quota, the domestic price equals the world price. Consumer surplus, the area between the demand curve and the world price, is area $A + B + C + D + E' + E'' + F$. Producer surplus, the area between the supply curve and the world price, is area G . The surplus of licence holders equals zero because there are no licences. Total surplus, the sum of consumer, producer and licence holder surplus, is area $A + B + C + D + E' + E'' + F + G$.

After the government imposes the import quota and issues the licences, the domestic price exceeds the world price. Domestic consumers get surplus equal to area $A + B$, and domestic producers get surplus equal to area $C + G$. The licence holders make a profit on each unit imported equal to the difference between the Isolandian price of olive oil and the world price. Their surplus equals this price differential times the quantity of imports, equal to the area of the rectangle $E' + E''$. Total surplus with the quota is the area $A + B + C + E' + E'' + G$.

To see how total welfare changes with the imposition of the quota, we add the change in consumer surplus (which is negative), the change in producer surplus (positive) and the change in licence holder surplus (positive). We find that total surplus in the market decreases by the area $D + F$. This area represents the deadweight loss of the import quota.

This analysis should seem somewhat familiar. Indeed, if you compare the analysis of import quotas in Figure 17.13 with the analysis of tariffs in Figure 17.12, you will see that they are essentially identical. Both tariffs and import quotas raise the domestic price of the good, reduce the welfare of domestic consumers, increase the welfare of domestic producers and cause deadweight losses. There is only one difference between these two types of trade restriction: a tariff raises revenue for the government (area E in Figure 17.12), whereas an import quota creates surplus for licence holders (area $E' + E''$ in Figure 17.13).

Tariffs and import quotas can be made to look even more similar. Suppose that the government tries to capture the licence holder surplus for itself by charging a fee for the licences. A licence to sell 1 tonne of olive oil is worth exactly the difference between the Isolandian price of olive oil and the world price, and the government can set the licence fee as high as this price differential. If the government does this, the licence fee for imports works exactly like a tariff: consumer surplus, producer surplus and government revenue are exactly the same under the two policies.

In practice, however, countries that restrict trade with import quotas rarely do so by selling the import licences. For example, in 1991, the EU reached an agreement with Japan to 'voluntarily' limit the sale of Japanese cars in member countries of the EU. In this case, the Japanese government allocates the import licences to Japanese firms, and the surplus from these licences (area $E' + E''$) accrues to those firms. This kind of import quota is, from the standpoint of the welfare of the EU, strictly worse than an EU tariff on imported cars. Both a tariff and an import quota raise prices, restrict trade and cause deadweight losses, but at least the tariff produces revenue for the EU rather than for Japanese auto companies. It is perhaps not surprising, therefore, that this arrangement was terminated at the end of 1999.

Although in our analysis so far import quotas and tariffs appear to cause similar deadweight losses, a quota can potentially cause an even larger deadweight loss, depending on the mechanism used to allocate the import licences. Suppose that when Isoland imposes a quota, everyone understands that the licences will go to those who spend the most resources lobbying the Isolandian government. In this case, there is an implicit licence fee – the cost of lobbying. The revenues from this fee, however, rather than being collected by the government, are spent on lobbying expenses. The deadweight losses from this type of quota include not only the losses from overproduction (area D) and under-consumption (area F) but also whatever part of the licence holder surplus (area $E' + E''$) is wasted on the cost of lobbying.

Non-Tariff Barriers

Barriers to trade are sometimes not obvious, but nevertheless present significant restrictions on the ability of firms to buy and sell goods from and to other countries. We will briefly outline some of the main ones.

Complex or Discriminatory Rules of Origin and Quality Conditions Countries may impose strict rules on the production of goods in its domestic market relating to technical specifications, health and safety, production standards and so on. Exporters may find it difficult to meet these rules or if they can, the rules increase the cost of production considerably and thus make the imports less competitive against domestically produced goods. In addition, firms may be required to give precise details as to where goods come from – something which is not easy given the widespread use of many different component parts from across the globe in many cases. The country establishing the non-tariff barriers (NTB) may only allow goods to be imported if they adhere to strict rules of origin, which exporters might not be able to meet.

Sanitary or Phyto Sanitary Conditions Firms may be required to provide details of food and plant exports, which again have to adhere to strict conditions. Phyto sanitary refers to the health of plants. Exporting firms may have to show that plants are free from pest and disease, and that measures have been taken to ensure the conditions of growth adhere to the standards laid down by the country setting the NTB. In the case of food exports, a country may set very high food safety standards and regulations which exporters find difficult or costly to meet.

Administrative Regulations Some countries might set up administrative procedures that must be met prior to any goods or services coming into the country. The paperwork or ‘red tape’ involved can be excessive and add to the costs of the exporter, which again makes the firm less competitive. Some countries might require excessive or unreasonable labelling, or packaging regulations, or set burdensome customs entry procedures, including the necessity of providing valuations of goods at the point of entry, which might be challenged (given value can be a subjective issue), or certification to prove authenticity, which again can lead to an increase in costs.

Currency Manipulation Some countries might implement measures to artificially influence the value of their currency, leading to exporters facing a higher price than would otherwise be the case, which reduces their competitiveness against domestic producers.

The Lessons for Trade Policy The team of Isolandian economists can now write to the new president:

Dear Madam President,

You asked us three questions about opening up trade. These are our answers.

Question: If the government allowed Isolandians to import and export olive oil, what would happen to the price of olive oil and the quantity of olive oil sold in the domestic olive oil market?

Answer: Once trade is allowed, the Isolandian price of olive oil would be driven to equal the price prevailing around the world. If the world price is higher than the Isolandian price, our price would rise. The higher price would reduce the amount of olive oil Isolandians consume and raise the amount of olive oil that Isolandians produce. Isoland would, therefore, become an olive oil exporter. This occurs because, in this case, Isoland would have a comparative advantage in producing olive oil.

Conversely, if the world price is lower than the Isolandian price, our price would fall. The lower price would raise the amount of olive oil that Isolandians consume and lower the amount of olive oil that Isolandians produce. Isoland would, therefore, become an olive oil importer. This occurs because, in this case, other countries would have a comparative advantage in producing olive oil.

Question: Who would gain from free trade in olive oil and who would lose, and would the gains exceed the losses?

Answer: The answer depends on whether the price rises or falls when trade is allowed. If the price rises, producers of olive oil gain and consumers of olive oil lose. If the price falls, consumers gain and producers lose. In both cases, the gains are larger than the losses. Thus, free trade raises the total welfare of Isolandians.

Question: Should a tariff or an import quota be part of the new trade policy?

Answer: A tariff, like most taxes, has deadweight losses: the revenue raised would be smaller than the losses to the buyers and sellers. In this case, the deadweight losses occur because the tariff would move the economy closer to our current no-trade equilibrium. An import quota works much like a tariff and would cause similar deadweight losses. The best policy, from the standpoint of economic efficiency, would be to allow trade without a tariff or an import quota.

We hope you find these answers helpful as you decide on your new policy.

Your obedient servants,
Isolandian economics team

SELF TEST Draw the supply and demand curve for wool suits in the country of Autarka. When trade is allowed, the price of a suit falls from 300 to 200 grams of gold. In your diagram, what is the change in consumer surplus, the change in producer surplus and the change in total surplus? How would a tariff on suit imports alter these effects?

The Arguments for Restricting Trade

The letter from the economics team persuades the new president of Isoland to consider opening up trade in olive oil. She notes that the domestic price is now high compared to the world price. Free trade would, therefore, cause the price of olive oil to fall and hurt domestic olive oil producers. Before implementing the new policy, she asks Isolandian olive oil companies to comment on the economists' advice.

Not surprisingly, the olive oil companies are opposed to free trade in olive oil. They believe that the government should protect the domestic olive oil industry from foreign competition. Let's consider some of the arguments they might give to support their position and how the economics team might respond.

The Jobs Argument Opponents of free trade often argue that trade with other countries destroys domestic jobs. In our example, free trade in olive oil would cause the price of olive oil to fall, reducing the quantity of olive oil produced in Isoland and thus reducing employment in the Isolandian olive oil industry. Some Isolandian olive oil workers would lose their jobs.

The counter argument to this is that trade can create jobs at the same time that it destroys them. When Isolandians buy olive oil from other countries, those countries obtain the resources to buy other goods from Isoland. Isolandian workers would move from the olive oil industry to those industries in which Isoland has a comparative advantage. This assumes that workers can move easily between different jobs, which of course, is possible but by no means cost free. The movement of workers between industries does impose hardship in the short run, but in the long run it can be argued that it allows Isolandians as a whole to enjoy a higher standard of living.

The National Security Argument When an industry is threatened with competition from other countries, opponents of free trade often argue that the industry is vital for national security. Free trade would allow Isoland to become dependent on foreign countries to supply vital resources. If a war later broke out, Isoland might be unable to produce enough to defend itself and remain self-sufficient.

Some economists acknowledge that protecting key industries may be appropriate when there are legitimate concerns over national security. It is also possible that this argument may be used too quickly by producers eager to gain at consumers' expense. Certainly, it is tempting for those in an industry to exaggerate their role in national defence to obtain protection from foreign competition.

The Infant Industry Argument New industries sometimes argue for temporary trade restrictions to help them get started. After a period of protection, the argument goes, these industries will mature and be able to compete with foreign competitors. Similarly, older industries sometimes argue that they need protection to help them adjust to new conditions. In 2002, for example, when US President George Bush imposed steep tariffs on the import of steel from the EU, he argued that the industry needed protection

in order to be able to afford to pay the pensions and healthcare costs of its retired workers, and while it was going through a period of adjustment to make its production more efficient in order to be able to cope with intense foreign competition.

Some economists are sceptical about such claims. The primary reason is that the infant industry argument is difficult to implement in practice. To apply protection successfully, the government would need to decide which industries will eventually be profitable and whether the benefits of establishing these industries exceed the costs to consumers of protection. Yet ‘picking winners’ is extraordinarily difficult. It is made even more difficult by the political process, which often awards protection to those industries that are politically powerful. Once a powerful industry is protected from foreign competition, the ‘temporary’ policy is hard to remove.

In addition, some economists are sceptical about the infant industry argument even in principle. Suppose, for instance, that the Isolandian olive oil industry is young and unable to compete profitably against foreign rivals, but there is reason to believe that the industry can be profitable in the long run. In this case, the owners of the firms should be willing to incur temporary losses to obtain the eventual profits. Protection is not necessary for an industry to grow. Firms in various industries – such as many internet firms today – incur temporary losses in the hope of growing and becoming profitable in the future. Many of them succeed, even without protection from foreign competition.

The Unfair Competition Argument A common argument is that free trade is desirable only if all countries play by the same rules. If firms in different countries are subject to different laws and regulations, then it is unfair (the argument goes) to expect the firms to compete in the international marketplace. For instance, suppose that the government of Neighbourland subsidizes its olive oil industry by giving olive oil companies large tax breaks. The Isolandian olive oil industry might argue that it should be protected from this foreign competition because Neighbourland is not competing fairly.

Would it, in fact, hurt Isoland to buy olive oil from another country at a subsidized price? Certainly, Isolandian olive oil producers would suffer, but Isolandian olive oil consumers would benefit from the low price. Moreover, the case for free trade is no different: the gains of the consumers from buying at the low price would exceed the losses of the producers. Neighbourland’s subsidy to its olive oil industry may be a bad policy, but it is the taxpayers of Neighbourland who bear the burden. Isoland can benefit from the opportunity to buy olive oil at a subsidized price.

The Protection as a Bargaining Chip Argument Another argument for trade restrictions concerns the strategy of bargaining. Many policymakers claim to support free trade but, at the same time, argue that trade restrictions can be useful when we bargain with our trading partners. They claim that the threat of a trade restriction can help remove a trade restriction already imposed by a foreign government. For example, Isoland might threaten to impose a tariff on olive oil unless Neighbourland removes its tariff on wheat. If Neighbourland responds to this threat by removing its tariff, the result can be freer trade.

The problem with this bargaining strategy is that the threat may not work. If it doesn’t work, the country has a difficult choice. It can carry out its threat and implement the trade restriction, which would reduce its own economic welfare. Or it can back down from its threat, which would cause it to lose prestige in international affairs. Faced with this choice, the country might wish that it had never made the threat in the first place.

CASE STUDY The US–China Trade War

For many years, some Americans have complained that trade with China is ‘unfair’. China has been accused of manipulating its currency to keep export prices artificially low and so gain an advantage in international trade. The election of Donald Trump with a promise to ‘Make America Great Again’ and put the interests of US workers and jobs first led to Mr Trump threatening to impose tariffs on some Chinese imports into the United States. That threat was countered by China with tariffs being imposed on US imports into China. In March 2018, the United States announced tariffs would be imposed on \$3 billion worth of Chinese goods coming into the United States. China retaliated by imposing \$3 billion worth of

tariffs on US exports in early April. This was met with a further response from the United States of \$46 billion worth of tariffs on Chinese imports on 2 April. China announced \$50 billion of tariffs on US imports on 4 April, and a day later, the United States announced tariffs on \$100 billion of Chinese imports.

The tit-for-tat threats and announcements continued into the summer, and in July 2018 the United States made good a threat originally made in June by imposing tariffs of 10 per cent on \$200 billion worth of Chinese imports. There was a wide range of goods included in the list subjected to the \$200bn, including construction materials, parts for cars, fish, rice, vegetables and even bovine semen.

A statement released by the US Government in July 2018, stated:

For many years, China has pursued abusive trading practices with regard to intellectual property and innovation. [US Trade Representative] (USTR) conducted a thorough investigation over an 8-month period, including public hearings and submissions. In a detailed 200-page report, USTR found that China has been engaging in industrial policy which has resulted in the transfer and theft of intellectual property and technology to the detriment of our economy and the future of our workers and businesses.

China rejected the claims made by the United States.

This 'trade war' did not have the universal support of US businesses. Some complained that retaliatory tariffs by China was damaging their business, and other businesses in the United States noted that the tariffs would increase import costs and affect their competitiveness. Supporters of tariffs might argue that higher import costs would force importers to turn to domestically produced products, but it is not always that simple. Globally, the trade dispute acted to depress stock markets as the prospect of a decline in global trade and the resulting effects on economic growth were calculated.



The trade dispute between the US and China has implications for the global economy.

CRITICISMS OF COMPARATIVE ADVANTAGE THEORY

The theory of comparative advantage has appeal and intuitively makes sense. In some surveys of economists, there seems to be a considerable number who agree that free trade is essentially a 'good' thing. The belief in free trade is based on the benefits espoused by the theory of comparative advantage. As noted earlier, there have been criticisms of the theory because of the context in which it was developed and that its assumptions are too simplistic to describe modern economies.

For example, countries with a large supply of unskilled labour and land might reasonably be expected to have a comparative advantage in the production of primary products and to trade the surplus of these with other more developed nations who may have a comparative advantage in the production of manufactured goods. In Africa, this scenario applies to many countries, but these countries seem to have failed to reap the benefits of trade in the way that the theory suggests and remain extremely poor.

Other developing economies like India, China and South Korea, in contrast, have seen more rapid growth and improvement in their citizens' well-being. China's leaders have invested heavily in manufacturing industry, South Korea in large enterprises, and India has a reputation in software and computer program development. Given that there are plenty of other countries in the world that had developed industries in these fields, it can be argued that India, China and South Korea were not exploiting comparative advantage. Rather, a conscious decision by the authorities to invest and build skills in industries where they did not have a comparative advantage has helped them to develop at a faster rate than many countries in Africa.

Rather than focus on comparative advantage, a more relevant point is that countries may make active decisions to specialize not in those industries in which it has the factor endowments that give them comparative advantage, but in industries where the benefits to the population as a whole are likely to be greatest. For countries like China, South Korea and India, these decisions might be based on what is called the **Prebisch–Singer hypothesis**. This hypothesis states that as incomes rise, spending on manufactured goods rises, while spending on primary products falls. For countries focusing on primary products, like many in Africa which have appropriate resource endowments, they are likely to become poorer over time compared to those which invest in manufactured goods production.

Prebisch–Singer hypothesis a hypothesis suggesting that the rate at which primary products exchange for manufactured goods declines over time, meaning that countries specializing in primary goods production become poorer

Attempts to impose free trade on countries based on the theory of comparative advantage, therefore, could lead to countries not gaining the benefits the theory predicts. Joan Robinson noted that Ricardo's analysis focused on Britain and Portugal and used cloth and wine as the two goods concerned, with Britain having the comparative advantage in cloth production and Portugal in wine production. In specializing in wine production, the cloth industry in Portugal withered, but the benefits from exporting wine were also limited because the global market for wine was not expanding quickly.

Empirical research into the Prebisch–Singer hypothesis, published in 2013 by Arezki, Hadri, Lougani and Rao for the International Monetary Fund, looked at 25 primary products since 1650 and found that results on the Prebisch–Singer hypothesis were mixed but that 'in the majority of cases, the Prebisch–Singer hypothesis is not rejected'. Critics of the theory of comparative advantage thus point to historical evidence that gains from trade can be garnered through investment in goods that have higher value in world trade markets rather than in goods where resource endowments suggest they have a comparative advantage.

OTHER THEORIES OF INTERNATIONAL TRADE

There have been a number of other theories of international trade which have been developed and we will provide an overview of some of these in the next section.

Factor Endowments: The Hecksher–Ohlin Theory

This theory derives from work by two Swedish economists, Eli Hecksher and Bertil Ohlin, the latter being jointly awarded the Nobel Prize in Economics in 1979 (with James Meade) for his work on the theory of international trade.

Hecksher and Ohlin's theory was based around the idea that any produced good or service represents a bundle of factors of production which go into their production. Different countries have different endowments of factors of production. Some have abundant supplies of raw materials such as oil, copper, iron ore and so on, whereas others might have greater supplies of productive land and labour. This means that different countries will have a comparative advantage in the production of goods and services in which they have relevant factor endowments. Saudi Arabia, for example, has a comparative advantage in the production of oil, given the large reserves of oil it has, whereas Bangladesh has a comparative advantage in the production of goods which utilize labour-intensive production methods.

Of course, most countries will import raw materials, component parts and semi-finished products and use these to help produce goods and services which they then export. Countries, therefore, have different patterns of trade depending on the relative proportions of imports and exports. This pattern of trade will be partly dependent on the relative factor endowments that countries have. The Heckscher–Ohlin theory offers a prediction on what this pattern of trade is likely to be.

In simple terms, it is likely that a country will tend to import goods and services where the factor endowments to produce those goods and services in that country are relatively scarce. Equally, where a country has a relative abundance of factor endowments which enable it to produce goods and services, it is likely to export those goods and services. There are benefits to trade, therefore, between countries with different factor endowments.

For example, assume two countries which use only capital and labour in the production of two goods, X and Y. Country A is a capital abundant country, having a greater factor endowment of capital in relation to labour, whereas country B is endowed with a greater amount of labour in relation to capital. The amount of capital per unit of labour is higher in country A whereas the amount of labour per unit of capital is greater in country B. If we look at the two goods, assume that the production of good X requires more units of capital per unit of labour to produce. Good X, therefore, is relatively capital intensive. Good Y, on the other hand, requires more labour per unit of capital in production and so is the more labour intensive good.

The Heckscher–Ohlin theory predicts that country A would produce good X, the capital intensive good, and country B would produce good Y, the labour intensive good. Country A would export good X and import good Y, and country B would export good Y and import good X. Both countries benefit from trade. In summary, the theory predicts that a country will export goods that are intensive in the factor in which it has an abundance and import goods which are intensive in the factor in which its factor endowment is, in relative terms, more limited.

Patterns of trade can change as a result of changes in relative factor endowments. In China, for example, the existence of large quantities of relatively cheap labour meant that China was able to export goods which were relatively labour intensive. In recent years, the innovation and investment in capital in China is shifting the relative factor endowments of the country and its trading pattern is changing as a consequence. These factor endowments, therefore, are not set and governments can adjust endowments through judicious investment and incentives, as in the cases of India and South Korea.

The Stolper–Samuelson Theory

The Heckscher–Ohlin model is based on two factors of production. Within this model, Paul Samuelson and Wolfgang Stolper explored what might happen if output and prices in a two-factor model changed as a result of trade. In our analysis of international trade, it has been suggested that free trade can benefit countries as a whole, even though within countries involved in trade there will be individuals who are winners and some who are losers. Over the years, attempts to negotiate a reduction in global tariff barriers and open economies of both the developed world and the less developed world to trade has been a focus of global economic policy. However, if economies of low-wage countries enter into trade with economies of high-wage countries, would those workers in high-wage economies suffer?

Intuitively, it might seem obvious that free trade could indeed affect workers in high-wage economies. Some economists would reject this intuition on the basis that low wages reflect the fact that workers are not as productive. In ‘poorer’ countries, the supply of large numbers of low-paid workers might be explained by the fact that investment in capital was low in comparison to high-wage economies and thus workers were nowhere near as productive.

We have seen how the Heckscher–Ohlin model suggested that a country will export goods that are intensive in the factor in which it has an abundance and import goods which are intensive in the factor in which its factor endowment is more limited. There will be winners and losers in international trade, but workers can move from industries where demand is declining in a country where cheaper imported goods replace domestically produced products, to those industries which are expanding because of the increase in export opportunities. Wolfgang Stolper and Paul Samuelson explored whether this would indeed be the case and investigated links between trade, tariffs and factor prices and developed the Stolper–Samuelson theorem.

The basis of the theory works as follows. Assume Isoland produces two goods, X and Y, using only the factor inputs land and labour. Good X is labour intensive and good Y is land intensive. A free trade agreement is made with another country which results in imports of good X into Isoland. This results in the supply of good X rising and the price of good X in Isoland falling. If the price of good X falls, then the wages paid to workers producing good X fall and some workers will find themselves out of work.

Workers producing good Y, the land intensive good, will also see a fall in their wages, because the supply of labour in Isoland has risen as a result of the workers in good X being made unemployed. The price of good Y will also fall, therefore. A fall in the price of good Y will result in an increase in quantity demanded, and businesses producing good Y seek to increase the amount of good Y being produced.

As good Y is land intensive, the increase in production of good Y leads to an increase in demand for land, which bids up its price. The result is that workers in Isoland face a fall in wages and an increase in the price they pay for land. The opening up of trade has had a damaging effect on workers in all industries.

This simple model can be extended to the way in which workers in developed countries have been affected as international trade has grown and trade has increased between developed economies and less developed economies. Labour in developed economies do not, based on this theory, benefit from international trade in quite the same way that traditional trade theory predicts.

Now take a further example. Isoland produces two goods, X and Y, with X being labour intensive as before, but good Y is now capital intensive. The wage rate paid to labour is W and the factor income to capital is R . Isoland not only produces the labour intensive good X, it also imports the good. Isoland exports quantities of the capital intensive good Y.

Assume that Isoland imposes a tariff on imports of good X. The price of good X increases as a result, and there is an incentive for producers of good X to increase output in response to the increase in price of good X. If we assume that all the factor inputs, labour and capital, are fully employed in producing goods X and Y in Isoland, the expansion in output of good X must come at the expense of good Y, which will see a contraction.

The expansion of output in good X leads to a rise in demand for labour (recall that labour is a derived demand), but the demand for capital remains stable. Higher wages in the good Y industry result in a reduction in the return on capital (R). Stolper and Samuelson hypothesized that the reduction in the return to capital would result in an increase in the wage rate of labour (W) that would be likely to be higher than the increase in the price of imports of good X. Their conclusion was that in the case where a country produces goods which compete with imports and which are relatively labour intensive (in this example), labour will benefit, but the owners of capital suffer from the effect of the tariff.

The Stolper–Samuelson theory predicts that when economies engage in international trade and face fixed world prices, there will be overall gains from a reduction in tariffs. Regardless, changes in trade policy will see both winners and losers. One of the implications of the Stolper–Samuelson theory is the extent to which the process of globalization, the increasing interdependence among world economies, has resulted in an increase in trade between low-wage economies and more developed economies, and increased wage differentials between low-skilled and higher-skilled wages. In studies carried out since the publication of the Stolper–Samuelson theory in the early 1940s, there has been some corroboration of the predictions made by the theory. However, there are equally studies which have cast doubt on the extent to which the theories assumptions and predictions hold.

The Imitation Gap or Imitation Lag Hypothesis

Countries may have different factor endowments, but dynamic events in the economy can result in one country innovating at a different rate from others. When new products are developed, it may take time before these products are replicated and/or developed further by other countries, hence the term ‘gap’ or ‘lag’ in this hypothesis. This hypothesis was developed by Michael V. Posner (1931–2006), a British economist who worked at the University of Cambridge and was also an advisor to the UK government.

When a new product is developed, the innovation may be centred in a particular country. This country, therefore, has an advantage in the production of this good. Assuming it is successful, there will be a period of time before rival firms are able to also produce variants of the same good and compete. The time lag will be dependent on the nature of the technology involved, the resources needed and the investment required to enable the product to be commercially viable and available.

In addition to the supply-side of the equation, there will also be demand-side lags. It might take time for the product to become known and accepted. If and when an alternative supply does become available from other countries, consumers in different countries may take time to come to accept and embrace the substitute products – if indeed they do at all. For example, Apple has introduced a number of innovative products which have resulted in ‘imitations’, most notably in smartphones and the associated operating systems which run applications. Other technology companies have attempted to replicate Apple products and developed rival mobile operating systems such as Android (developed by Google), Windows Phone, BlackBerry 10 and Ubuntu Touch. Some of these systems have been accepted by the market and others have not and have been discontinued. Even if new products are introduced which are very good imitations, it may be more challenging to gain market penetration because of existing brand loyalty and familiarity.

The imitation lag hypothesis, therefore, suggests that international trade and the pattern of imports and exports in different countries can be affected by the development of new products and highlights the importance of innovation to improving and maintaining export performance. Economic development can be driven by export-led growth and the imitation lag hypothesis points to the importance for countries of the need to support and encourage innovation, research and development, and entrepreneurship if the country is to maintain its position in international trade.

Product Cycle Theory

The product life cycle is a schematic representation of the stages a product goes through from development, through launch, growth, maturity and eventual decline. The basic principles of the product life cycle can be extended to international trade theory. Product cycle theory has become an extension of the imitation lag hypothesis in explaining changes in trade patterns. Product cycle theory was developed by the US economist, Raymond Vernon (1913–99). Product life cycle theory suggests that products are developed and go through an early growth stage (assuming they are successful). Product cycle theory notes that at this stage in the development of a product, it is most likely that sales will take place in the country in which the product was developed and so international trade in the product will be non-existent.

As the product enters the maturity stage, techniques of mass production and commercialization of the product will have been developed, and firms producing the product will be able to benefit from economies of scale. As the product becomes more competitive, it will begin to generate sales abroad as firms enter negotiations to develop trading agreements and contracts. If the product is one introduced in an already developed economy, it is likely that the trading opportunities will be with other similarly developed economies. However, it is also likely that less developed economies will begin copying these products, and if they have cheaper labour costs, for example, they can offer similar products for sale around the world at lower prices than those which may be in force in the country of origin. Trade patterns thus begin to change as countries with advantages of certain factor endowments begin to take market share. The domestic country in which the product originated might begin to find that exports of the product begin to slow, and even decline, while imports of the imitation products grow and gain greater market share.

As the product continues along the product life cycle (a stage Vernon referred to as the ‘standardised product stage’), production may have transferred almost entirely to developing countries, and developed economies will have shifted their focus onto developing other new products which help generate export revenue and growth. Product cycle theory reflects the dynamics of international trade and changing trade patterns.

Consumer Tastes: The Linder Theory

Countries in the world can be classified as ‘high income’, ‘middle income’ and ‘low income’. Consumers in these different countries will have different purchasing habits and patterns, and this can be reflected in the types of goods and services produced in the country. For example, in a high-income country, the type of goods and services domestic firms produce might cater for more sophisticated financial products, more technology based luxuries and more highly processed foods. In low-income countries, these types of goods would not be in such high demand. Firms in these countries would tend to produce goods and services which might be regarded as basic necessities. As a consequence, the trade which different countries engage in is a function of the goods and services it produces.

Swedish economist, Hans Martin Staffan Burenstam Linder (1931–2000), argued that trade patterns are influenced by the extent to which there is an overlap in the demand from consumers across countries with different income levels. In the theory of comparative advantage, we stated that it was not possible for a country to have a comparative advantage in all goods, because the opportunity cost of one good is the inverse of the opportunity cost of another. The Linder hypothesis argues that trade can take place in the same good and be beneficial to both countries provided consumer tastes overlap. This is observed in the UK, which both imports and exports oil. Linder argued that even if countries have different income levels, there can be instances where there is a common demand for goods, which can mean trade in those goods can take place and be beneficial to both countries.

CONCLUSION

The principle of comparative advantage shows that trade can make everyone better off. Despite these apparent benefits, there are fierce debates about trade policy. The theory of comparative advantage has been criticized as having assumptions which are too static and which do not reflect the way in which economies have developed since the original theory was developed in the early nineteenth century. Other theories of international trade have been developed over the years, but all are subject to debate and criticism.

Regardless of the benefits of free trade, most countries in the world have some sort of trade restrictions in place, or group together to get the benefits of free trade between themselves while imposing trade restrictions on those countries outside the free trade 'club'.

SUMMARY

- PPFs provide a model to show potential output of goods and services in an economy and the opportunity cost ratios of diverting resources to different uses.
- The PPF can shift outwards if countries find ways of improving their factor productivity or exploit factor endowments more effectively.
- The shape and position of a PPF is dependent on the productivity of factor inputs and degree of specialization involved in the country in different industries.
- Each person consumes goods and services produced by many other people, both in their country and around the world. Interdependence and trade are desirable because they allow people to enjoy a greater quantity and variety of goods and services.
- There are two ways to compare the ability of two people in producing a good. The person who can produce the good with the smaller quantity of inputs is said to have an *absolute advantage* in producing the good. The person who has the smaller opportunity cost of producing the good is said to have a *comparative advantage*. The gains from trade are based on comparative advantage, not absolute advantage.
- The theory of comparative advantage shows that trade can make everyone better off through specialization.
- The effects of free trade can be determined by comparing the domestic price without trade to the world price. A low domestic price indicates that the country has a comparative advantage in producing the good and that the country will become an exporter. A high domestic price indicates that the rest of the world has a comparative advantage in producing the good and that the country will become an importer.
- When a country engages in trade there are winners and losers, but many economists argue that the benefits to the winners outweigh the losses to the losers.
- Despite the supposed gains from trade, most countries impose some sort of barrier to trade. Two of the most commonly used barriers are tariffs and quotas.
- A tariff is a tax on imports and raises the price of imports coming into a country. Domestic producers of goods where a tariff has been imposed are better off, and the government raises revenue. However, consumers will be worse off.

- An import quota – a limit on imports – has effects that are similar to those of a tariff. Under a quota, however, the holders of the import licences receive the revenue that the government would collect from a tariff.
- There are various arguments for restricting trade: protecting jobs, defending national security, helping infant industries, preventing unfair competition and responding to foreign trade restrictions.
- Critics of the theory of comparative advantage argue that the assumptions of the theory do not hold in real life, that the development of the theory was based on a particular historical context which does not exist anymore, and that developing countries investing in higher value industries, rather than those in which they have the factor endowments which provide comparative advantage, have seen greater economic benefits.
- Other theories of international trade focus on factor endowments, take into account time lags in responding to a dynamic economy, the product cycle and changing consumer tastes.

IN THE NEWS



Trade Wars

The rallying cry of Donald Trump in the US presidential election in 2016 was to 'Make America Great Again' and to focus on policies that protected and boosted US jobs and US workers. On election, Mr Trump imposed a range of tariffs on goods like steel and aluminium and in 2018 a spate of tariffs was imposed on a range of Chinese goods amid claims from the United States that China did not 'play fair' in international trade, a claim which China rejects.

The move towards an increase in protectionism led to fears that global trade would contract affecting the growth potential for many countries. In the United States, the imposition of tariffs had the support of many people who claimed that the benefits of free trade were far more illusory than economists might suggest. As has been noted in this chapter, there will inevitably be winners and losers in any issue relating to international trade. What is important to consider is who are the winners, who are the losers, what power does each have and what is the extent of the gains and losses they suffer?

If the United States imposes a 10 per cent or even a 25 per cent tariff on the import of rice from China, for example, economics tells us that the demand for rice in the United States will fall as its price rises. That reduces the quantity of imports into the United States. But consider users of rice; does this mean they will stop using and consuming rice in the same quantities as before the tariff or will they seek to buy rice from elsewhere? One option would be to switch to buying rice from US producers. Rice is produced in parts of the United States, primarily California, Texas, Mississippi, Missouri and Arkansas, but US rice production only accounts for a small percentage of global production. Global production of rice is around 480 million tonnes whereas total production in the United States is only around 13 million tonnes, about 2.7 per cent. It is unlikely, therefore, that consumers of rice in the United States can switch to buying domestically produced rice. Unless consumption of rice falls dramatically, it is likely that imports of rice from China, which has the tariff, would switch to imports of rice from other countries which do not have tariffs imposed on them.

Existing rice producers like Vietnam, Malaysia and India might be the countries to step in and take advantage of the trade dispute between the United States and China and



The trade dispute between the US and China might provide opportunities for existing rice producers.

export their rice to the lucrative US market. Which of these countries is best placed to replace the Chinese exports of rice? That largely depends on how geared up producers are in the country to expand production and fill the gap left by China. That is not easy, and the institutional, legal and regulatory framework for business in each country will make it easier for some to respond than others. This in part is measured by what is called the 'Ease of Doing Business Index' (EoDBI). This index is compiled at the World Bank and is a measure of the legal, regulatory and institutional factors that exist in a country which influence the ease with which businesses can set up, operate and flourish.

In 2018, Malaysia was ranked 15th and classed as 'very easy', Vietnam came in at 69th classed as 'easy', and India was ranked 77th, classed as 'medium'. The EoDBI generates its rankings based on 10 key factors which include the tax regime, registering property, enforcing contracts, accessing credit and electricity, and the process which must be followed to set up a business. If countries like India, for example, want to be a 'winner' out of the trade dispute between the United States and China, its government might find it wise to focus on improving the comparative advantage India has over rivals like Malaysia and Vietnam.

Critical Thinking Questions

- 1** 'Protectionism is the only way in which a country can ensure that its own workers and jobs are secured from unfair competition from abroad.' Unpick this statement using the theories outlined in this chapter and comment on the extent to which you agree with the view in the statement.
- 2** Why do you think that many economists are supporters of free trade? Why might non-economists consider the benefits of trade to be 'illusory'?
- 3** Analyze the difference in the possible outcomes of a country imposing a tariff of 25 per cent on goods from one country to that of a similar tariff on imports of a good from a range of countries.
- 4** To what extent do you think the ease of doing business in a country can be a source of comparative advantage?
- 5** Assume that the United States imposes a tariff on rice exported by China. Analyze the economics of this decision taking into account different theories of international trade covered in this chapter.

QUESTIONS FOR REVIEW

- 1** Draw a PPF for a country producing only computers and wheat. What determines the shape and position of the PPF you have drawn? Use your diagram to show the opportunity cost of different output combinations.
- 2** What does the domestic price that prevails without international trade tell us about a nation's comparative advantage?
- 3** Explain how absolute advantage and comparative advantage differ.
- 4** Can two countries gain from trade if the opportunity cost ratios relating to the production of goods they can both produce is the same? Explain.
- 5** Thousands of tourists visit European cities every year. Is tourism an import or an export to a country? Explain.
- 6** Draw the supply and demand diagram for an importing country. What is consumer surplus and producer surplus before trade is allowed? What is consumer surplus and producer surplus with free trade? What is the change in total surplus?
- 7** Describe what a tariff is and its economic effects. What is an import quota? Compare its economic effects with those of a tariff.
- 8** List five arguments often given to support trade restrictions. How do economists respond to these arguments?
- 9** Outline the main principles of the role of factor endowments in the Heckscher–Ohlin theory.
- 10** How do time and the product cycle influence trade patterns?

PROBLEMS AND APPLICATIONS

- Draw the PPF for a country which produces just oranges and cars. Assume that the country is better at producing oranges than cars.
 - What can you say about the opportunity cost of the country diverting more resources into producing cars?
 - The country discovers new resources which will mean that its ability to produce cars is significantly improved. What happens to the shape of the PPF as a result of this discovery?
 - What can you say about the opportunity cost facing the country after this new discovery?
- Look at the table below which shows the production possibilities for two products, pens and pencils.

Point	Pens	Pencils
A	10	0
B	5	10
C	2	25

- Calculate the opportunity cost of producing 1 additional pen between points A and B.
 - Calculate the opportunity cost of producing 1 additional pencil between points A and B.
 - Calculate the opportunity cost of producing 1 additional pen between points B and C. What conclusions can you come to about this opportunity cost compared to the calculation you made in a. above?
 - Calculate the opportunity cost of producing 1 additional pencil between points B and C. What conclusions can you come to about this opportunity cost compared to the calculation you made in b. above?
- European and Chinese workers can each produce 4 capital goods a year. A European worker can produce 10 tonnes of grain a year, whereas a Chinese worker can produce 5 tonnes of grain a year. To keep things simple, assume that Europe and China have 100 million workers each.
 - For this situation, construct a table analogous to Table 17.1.
 - Graph the PPF of the European and Chinese economies.
 - For Europe, what is the opportunity cost of a car? Of grain? For China, what is the opportunity cost of a car? Of grain? Put this information in a table analogous to Table 17.3.
 - Which has an absolute advantage in producing capital goods? In producing grain?
 - Which has a comparative advantage in producing capital goods? In producing grain?
 - Without trade, half of Europe and China's workers produce capital goods and half produce grain. What quantities of capital goods and grain do Europe and China produce?
 - Starting from a position without trade, give an example in which trade makes both Europe and China better off.
 - The UK and Poland both produce cakes and coats. Suppose that a UK worker can produce 50 cakes per hour or 1 coat per hour. Suppose that a Polish worker can produce 40 cakes per hour or 2 coats per hour.
 - Which country has the absolute advantage in the production of each good? Which country has the comparative advantage?
 - If the UK and Poland decide to trade, which commodity will Poland trade to the UK? Explain.
 - If a Polish worker could produce only 1 coat per hour, would Poland still gain from trade? Would the UK still gain from trade? Explain.
 - The world price of wine is below the price that would prevail in France in the absence of trade.
 - Assuming that French imports of wine are a small part of total world wine production, draw a graph for the French market for wine under free trade. Identify consumer surplus, producer surplus and total surplus in an appropriate table.
 - Now suppose that an outbreak of phyloxera (a sap sucking insect which damages grape vines) in California and South America destroys much of the grape harvest there. What effect does this shock have on the world price of wine? Using your graph and table from part (a), show the effect on consumer surplus, producer surplus and total surplus in France. Who are the winners and losers? Is France better or worse off?
 - Suppose that EU countries impose a common tariff on imported cars to protect the European car industry from foreign competition. Assuming that Europe is a price-taker in the world car market, show on a diagram: the change in the quantity of imports, the loss to European consumers, the gain to European car manufacturers, government revenue and the deadweight loss associated with the tariff. The loss to consumers can be decomposed into three pieces: a transfer to domestic producers, a transfer to the government and a deadweight loss. Use your diagram to identify these three pieces.

- 7** Answer the questions on comparative advantage below.
- Explain how countries in Africa, which have an abundance of labour as a natural resource, can benefit from trade and allow countries to grow by specializing in the export of primary products.
 - What factors might have caused countries in Africa to have not exploited their comparative advantage so that they have remained relatively poor in comparison to some of the more rapidly developed nations?
 - Does the Prebisch–Singer hypothesis suggest that the theory of comparative advantage is of limited use in explaining modern trade between nations?
- 8** Explain why the increase in globalization since the late 1960s might have resulted in a widening of the gap between the pay of low-skilled workers and high-skilled workers in developed economies.
- 9** Innovators in a country develop new technology which revolutionizes the way minerals can be extracted in mining industries, reducing the capital cost considerably. How would you expect this innovation will affect patterns of trade in subsequent years across the globe?
- 10** To what extent do you think there is a link between the product cycle theory as developed by Vernon and the Linder theory? Explain.